

1.5T HD Head, Neck, and Spine Array and 3.0T Head, Neck, and Spine Array Coils

Operator Manual
2423588-1EN



Operator Manual for the 1.5T HD Head,
Neck, and Spine Array and the 3.0T
Head, Neck, and Spine Array Coils
2423588-1EN
Revision 5 (2025-02)



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Revision History

Table i: Revision History

Revision	Date	Author	Reason for Change
01	2013-Apr-17	D Seaman	Initial Release
02	2016-Apr-21	D Seaman	- Title - updated rev and date, removed 'doing business as', removed GE Healthcare - all - updated rev, date and General Electric Company in footer - i - added usa to mfg address - ii - updated Intended use for the 1.5T hns coil to match the 510(k) intended use section - TOC - updated to match contents 1-1 - removed 'healthcare' 1-2, 1-3 - updated HNS part name in shipping tables to match 5735981 UDI serial label 1-3 - added 1.5T Silent HNS shipping table 1-4, 1-5 - updated HNS part name in shipping tables to match 5721177 UDI serial label 1-9 - removed text from under fragile, rain, and up images 1-10 - removed image of temp, humidity, pressure label; added images and definitions of individual temp, humidity and pressure symbols 2-4 - standardized 'looped cables' caution 2-7 - removed 'healthcare' 4-1 - removed 'healthcare' 4-2 - separated coil maintenance and coil cleaning sections 4-3 - added General Cleaning and Disinfection section 4-4 - removed 'healthcare', updated Accessories table to match HNS components found on GE Healthcare Service shop website
03	2018-May-07	D Seaman	- Throughout - updated revision, revision date and copyright date, changed caution symbol background color - title - update revision, revision date, copyright year, add CE mark, spelled out revision in lower right of page - i - added revision table - iii - added year of CE marking statement - iv - moved Environmental Requirements and Coil Storage to page iv - v - moved contact information to page v, added Electronic Instructions for Use section - TOC - updated to match updated contents 1-2 - added 2nd part number to row 3 of HNS coil 2418627 and 2418626 tables 1-8 - added second receive symbol, added caution symbol 2-1 - moved Intended use statements to page 2-1 3-12 - removed “#” symbols from Figure 3-15 3-19 - added “Please refer back...” sentence to Using Configurations section 3-20 to 3-42 - removed untranslatable text from Element Group images, anonymized images, spelled out "Posterior" where needed in the Element Groups row at the top of the page, added "Example Image" to cells containing the example image 3-44 - added "Please refer back..." sentence to For Users of the MR750w and Newer Systems section 3-44 to 3-53 - removed untranslatable text from Element Group images, anonymized images, added Group name and numbers to Scan Area cells 3-56 - anonymized photograph at step 3 3-57 - anonymized photograph at step 6 4-1 - removed phone number from Quality Assurance paragraph two 4-3 - removed phone number from "If you do not..." sentence in the Accessories and Replaceable Components section

Table i: Revision History (Continued)

Revision	Date	Author	Reason for Change
04	2023-01	D Seaman	• Throughout – updated fonts, revision, revision date, copyright year, company name • Title page – removed notified body number from CE symbol • Page i – viii – updated table headings • Page iii – changed to Medical Device Regulation section, removed notified body number from CE symbol, added EC/Rep and CH/Rep symbols, added EU Authorized Representative Heading, capitalized the “R” in Rue, added Swiss Representative Section • Page iv – added trashcan without the black bar to “This symbol...” caution, added “All packages...” caution, added Radio Frequency Energy Information section • Page v – updated Electronic Instructions for Use section • TOC – updated to match manual contents • Page 1-1 – replaced the “#” symbol with the word “number” in the 1st note and removed the “#” symbol from the 2nd note • Page 1-2 – 1-15 – reformatted table title numbers • Page 1-2 to 1-5 – added table numbers and replaced the number symbol with the words “part number”, changed “qty” to “Quantity” in table headings • Page 1-2 – 1-4 – replaced the shortened “sys” with “system” in tables 1-1, 1-2, 1-3, 1-6 • Page 1-4 – replaced shortened “cbl” with “cable” in table 1-5 and 1-6 and shortened “assy” with “assembly” in table 1-5 • Page 1-6 – added Important symbol and definition • Page 1-8 – added trashcan without black bar, receive symbol with R in lower right corner in the symbol column, added note to definition column that receive symbol may be in color or black and white, added element indicator symbol row • Page 1-9 – added FRU, Field, Ref Rev, eIFU, locked symbol, unlocked symbol, 1.5T, and 3.0T rows • Page 1-10 – added A, B, MD, P-port plug, P-port do not plug, P-port system dependent, and EC/Rep symbol rows • Page 1-11 – added CH/Rep and transmission symbol rows, removed “packaging” from CE symbol definition • Page 1-14 – added Clinical Benefits of MR Coils section • Page 1-15 – added Coil Specifications table heading, updated “#” symbol to the word “number” in table heading • Page 2-2 – added contraindications section, updated “RF can cause localized heating...” warning by combining it with the “Closed loops...” warning then removing the “Closed loops...” warning • Page 2-7 – added element indicator label • Page 2-8 – added “For non-p-connector HNS coils connector symbols label locations and definitions • Page 2-9 – added “For HNS coils with P-connectors label locations and definitions • Page 2-10 – updated Emergency Procedures and Incident Reporting section by calling out new headings and adding the Reporting of Serious Incidents section, changed “#” symbol to the word “number” in 3rd bullet point in “If a patient...” section • Page 2-11 – changed “#” symbol to the word “number” in 1st paragraph under the MR Safety Information heading • Page 3-1 – added Coil Installation section • Page 4-1 – added “Please contact a service engineer if:” and added bullet points to Coil Maintenance section • Page 4-2 – added “clean and disinfect the device after removing from packaging...” sentence and updated remainder of statement, added important and caution symbols to “Inspect pads...” statement, removed “The HNS Coil and patient comfort pads...” sentence because instruction is covered by the “Clean and disinfect...” instruction • Page 4-3 – 4-4 – updated section heading and contents of the Cleaning and Disinfection Recommendation for the MR System, Coils and Most Accessories section • Added Glossary

Table i: Revision History (Continued)

Revision	Date	Author	Reason for Change
05	2025-02	D Seaman	• Updated @ge.com to @gehealthcare.com addresses.

Medical Device Regulation

This product conforms with the requirements of the Medical Device Regulation 2017/745 when they bear the following CE Mark of conformity:



The year of CE Marking is 2009.

EU Authorized Representative



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283 Rue de la Minière
78530 BUC France
Telephone: +33 1 30 70 40 40

Swiss Authorized Representative



GE Medical Systems (Schweiz) AG
Europa-Strasse 31
8152 Glattbrugg
Switzerland



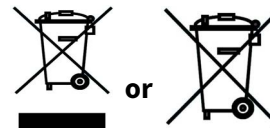
United States federal law restricts the sale, distribution and use of this device to, or on the order of, a physician.

CAUTION:



CAUTION:

This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.



CAUTION:

All packages should be examined at time of delivery. Please look for damaged packaging and inspect each coil component, cable and accessory to ensure there is no equipment damage from shipping. Visually check the coil for tears or damage. Ensure no internal conductors are exposed. If any coil component, cable, or accessory is suspected of not being in good condition, please consult a GE representative before use.

Radio Frequency Energy Information

This equipment generates, uses, and can radiate radio frequency energy. The equipment may cause radio frequency interference with other medical and non-medical devices and radio communications. To provide reasonable protection against such interference, the GE MR Systems comply with emissions limits for (Group 2, Class A) Medical Devices as stated in EN 60601-1-2. However, there is no guarantee that interference will not occur in a particular installation.



If this equipment is found to cause interference (which may be determined by turning the equipment on and off), the user (or qualified service personnel) should attempt to correct the problem by one or more of the following measure(s):

- Re-orient or relocate the affected device(s);
- Increase the separation between the equipment and the affected device;
- Power the equipment from a source different from that of the affected device and/ or
- Consult the point of purchase or service representative for further suggestions

The manufacturer is not responsible for any interference caused by using other than recommended interconnect cables or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the users' authority to operate the equipment.

Do not use devices, which intentionally transmit RF Signals (cellular phones, transceivers, or radio- controlled products) in the vicinity of this equipment as they may cause performance outside the published specifications. Keep the power to these types of devices turned off when near this equipment.

The medical staff in charge of this equipment is required to instruct technicians, patients, and other people who may be around this equipment to fully comply with the above requirement.

Immunity/Emissions Exceptions: Note the exceptions from the EMC test results. Check with the business EMC engineer for this information.

Environmental Requirements

Table ii: Environmental Requirements

Environmental Factor	Transport Conditions	Operating and Storage Conditions
Barometric pressure	645 mm Hg to 795 mm Hg (860 hPa to 1060 hPa)	As established by the GE recommended MR suite conditions.
Humidity	20% to 95% non-condensing	
Temperature	-15°C to 55°C (5°F to 131°F)	

Coil Storage

When the coil is not in transport and is not being used, store the coil in an air conditioned area that has environmental conditions that fall within the established GE recommended MR suite conditions.

To store the coil, a storage space that exceeds the dimensions of the coil is required (51 inches wide x 35 inches deep x 18 inches high) (130 cm wide x 89 cm deep x 46 cm high).

The coil should be stored in a horizontal position similar to its orientation on the MRI table. Head Neck Unit cables should be carefully placed through the two cable slots on the posterior and then the anterior should be latched in place. The TLU and NCU cables should be stored in a way that does not expose the cable to kinks or undue stress. The NCU and Anterior Adapter should be stored flat with no weight placed on top of them.

Contact Information

Table iii: Contact Information

GE Healthcare Americas	
USA	800-558-5102
Canada	800-668-0732
GE Healthcare Global	
China	86-21-62192228
Taiwan	886-2-2505-7900
Singapore	65-291-8528
Australia	61-2-9975-5501
Japan	81-120-48-2630
Korea	82-31-740-6119
India	91-80-845-2923
Europe	33-1-41-19-76-76
GE Website	www.gehealthcare.com
GE Healthcare Manufacturer Information	
Manufacturer Address	GE Healthcare Coils 1515 Danner Drive Aurora, Ohio 44202-9273 USA
Manufacturer Phone Number	1-330-562-1000

Electronic Instructions for Use



Your MR device is delivered with the electronic Instructions For Use (eIFU) logo. The electronic Instructions For Use (eIFU) symbol means that the operator manual can be viewed from a PC. Use these steps to view your eIFU manual from a personal computer.

1. Insert the multi-language DVD into your personal computer's CD-ROM drive. The minimum PC requirements include the following:
 - Pentium 90 MHz
 - Windows 95
 - 32 MB of RAM

- CD-ROM drive
 - Adobe Acrobat Reader 4.05 or higher. If your personal computer does not have Acrobat Reader, you can install it directly from the operator manual DVD, or you can download it for free from the Adobe website (www.adobe.com).
2. Typically, the DVD will auto start. If not, Double-click the My Computer icon.
 3. Double-click the CD or DVD icon.
 4. Double-click the desired language to open the pdf file.

Refer e-User Leaflet 5452943-199, 5851600-199, or the e-User leaflet that came with your MR system for further instructions.

A paper copy of Instructions For Use can be ordered at no additional cost. Please send a request to your Sales or Service representative. They will transfer your request to gehc_manuals@gehealthcare.com. In application of EU Commission Regulations on electronic instructions for use of medical devices, in the European Union, your request should be processed within 7 days.

To obtain the latest version of the Instructions For Use, go to:

<https://www.gehealthcare.com/documentationlibrary> - and follow the instructions.

On the search page of the Customer Documentation Portal, enter 2423588-399 in the search field, and launch the search.

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Chapter 1

Introduction and Shipping Contents

Introduction

This manual describes the safety precautions, features, use and care of the 1.5T HNS Coil compatible with 1.5T GE MRI Systems with P connectors, HD connectors and Legacy Bendix connector, and the 3.0T HNS Coil compatible with 3.0T GE MRI Systems with P connectors, HD connectors and 3.0T Legacy connector.

Please review this manual thoroughly before using the device for imaging. Refer to your MR scanner manual in conjunction with this manual. If you have questions or comments on this manual, or need assistance with the use of this product, please contact your GE representative.

This manual should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this manual before attempting to use this product.**

This equipment is for use by qualified personnel only.

NOTE: This operator manual does not apply to the 1.5T HD Head, Neck and Spine, Part number 2416329, revision 2.

NOTE: This operator manual does not apply to the 3.0T Head, Neck and Spine that is manufactured by Medrad and is made up of part numbers 5325953-2 (Head Neck Unit), 5325953-3 (Thoracic Lumbar Unit) and 5325953-4 (Neck Chest Unit).

Shipping Contents: 1.5T HNS - Legacy Bendix and HD Connector Coil (3 Connectors)

Table 1-1: HNS Coil Part Number 2418627

Part Name	Part Number	Quantity
HD Head Neck and Spine Array	2416329	1
HNS System Cable Excite III, 8ch	2423004/5582296	1
HNU System Cable Excite II, 8ch	2423006	1
Assembly Mirror	2423315	1
Pad Kit	2418081	1
Phantom Positioner	5344844	1
Operator Manual Packet	2423588-500	1

Shipping Contents: 1.5T HNS - HD Connector Coil (2 Connectors)

Table 1-2: HNS Coil Part Number 2418626

Part Name	Part Number	Quantity
HD Head Neck and Spine Array	2416329	1
HNU System Cable Excite III, 16ch	2423003/5582297	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Phantom Positioner	5344844	1
Operator Manual Packet	2423588-500	1

Shipping Contents: 1.5T HNS - HD and P Connector Coil (2 Connectors)

Table 1-3: HNS Coil Part Number 2424381

Part Name	Part Number	Quantity
HD Head Neck and Spine Array	2416329	1
HNU System Cable P connector, 16ch	2423972	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Phantom Positioner	5344844	1
Operator Manual Packet	2423588-500	1

Shipping Contents: 1.5T Silent HNS - HD Connector Coil (2 Connectors)

Table 1-4: HNS Coil Part Number M5000AT

Part Name	Part Number	Quantity
HD Head Neck and Spine Array	2416329	1
Silent HNS Cable and Coil ID	5746247	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Phantom Positioner	5344844	1
Operator Manual Packet	2423588-500	1

Shipping Contents: 3.0T HNS Coil - 3.0T Legacy and HD Connector Coil (3 Connectors)

Table 1-5: HNS Coil Part Number M3335HD

Part Name	Part Number	Quantity
3.0T 16CH Head Neck Spine (HNS) Array	5341333	1
Assembly, 8CH EXC III HNU System Cable, 3.0T	5338551	1
System Cable, 8CH EXCITE II, HNU, 3.0T HNS	5344575	1
Assembly Mirror	2423315	1
Pad Kit	2418081	1
Phantom Positioner	5344844	1
Operator Manual Packet	2423588-500	1
Coil Definitions Disk	5325975	1
3.0T HNS Service Manual	5367722	1

Shipping Contents: 3.0T HNS Coil - HD Connector Coil (2 Connectors)

Table 1-6: HNS Coil Part Number M3335HC

Part Name	Part Number	Quantity
3.0T 16CH Head Neck Spine (HNS) Array	5341333	1
16CH EX III HNU System Cable, 3.0T	5338550	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Phantom Positioner	5344844	1
Operator Manual Packet	2423588-500	1
Coil Definitions Disk	5325975	1
3.0T HNS Service Manual	5367722	1

Shipping Contents: 3.0T HNS Coil - HD and P Connector Coil (2 Connectors)

Table 1-7: HNS Coil Part Number M7000AP

Part Name	Part Number	Quantity
3.0T 16CH Head Neck Spine (HNS) Array	5341333	1
System Cable, 16CH 3.0T HNS, P Connector	5338113	1
Assembly Mirror	2423315	1
Cable Exit Plug	2423029	1
Pad Kit	2418081	1
Phantom Positioner	5344844	1
Operator Manual Packet	2423588-500	1
Coil Definitions Disk	5325975	1
3.0T HNS Service Manual	5367722	1

Use of Symbols

Coil users should understand the following symbols used for caution and warning explanations:



Indicates a hazard that may cause a minor injury or property damage if instructions are not followed.

CAUTION:



Indicates a hazard that may cause severe personal injury, death, or substantial property damage if instructions are not followed.

WARNING:



Important indicates information where adherence to procedures is crucial or where your comprehension is necessary to apply a concept or effectively use the product.

Coil Classification

Table 1-8: Coil Classification

Symbol	Classification
	Class II equipment
	Type BF applied

- Suitable for continuous operation
- The IP Rating for the coil is IPX0. Ordinary equipment. Not rated against water ingress protection.
- Non-sterile device
- Equipment not suitable for use in the presence of a flammable anaesthetic mixture with air or with oxygen or nitrous oxide.
- The voltage rating of the coil is 10VDC maximum.

All surface areas of the TLU and HNU (with Anterior Unit or Adapter Block), except the bottom surface that sits on the patient table, are considered applied parts and may come into direct contact with the patient. All surface areas of the NCU are considered applied parts and may come into direct contact with the patient. Pads and/or thermal-resistant materials should be used to prevent the coil surfaces from touching the patient.

Pads and/or thermal-resistant material must be used to ensure the patient cannot touch the coil cables and connectors.

Symbol and Label Definition

Table 1-9: Symbol and Label Definition

Symbol	Definition
	Manufacturer
	Date of Manufacture
	Consult Accompanying Documents
	Consult Accompanying Documents
	Serial Number
	Part Number
	This label, affixed to the HNU Anterior and the Horseshoe, informs the user to firmly press down on all coil components to ensure that they are fully engaged and connected. Failure to do so may cause artifacts and decrease image quality.
	Do not insert connector with this side up. If connector is inserted with this side up, the coil will not operate.

Table 1-9: Symbol and Label Definition (Continued)

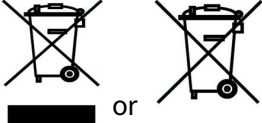
Symbol	Definition
	Do Not Loop the Cable Symbol
	Surface May Be Hot Symbol
 or 	This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.
 or  or 	Receive Only Coil Symbol. The Receive Only symbol may be in color or black and white.
	Pinch Point Warning Symbol
	Indicates a hazard that may cause a minor injury or property damage if instructions are not followed.
	Element Indicator label. When this label is on the coil, the “X” in the image to the left will be replaced by the actual number of the element.

Table 1-9: Symbol and Label Definition (Continued)

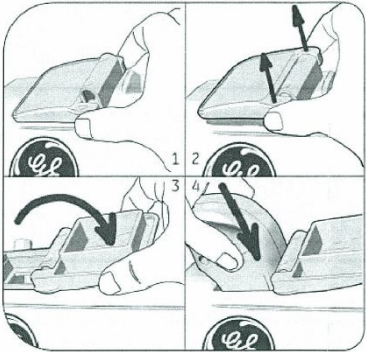
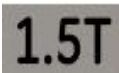
Symbol	Definition
	This label depicts how to properly lift and pivot the NCU connector cover to expose the connector and then plug in the NCU connector. The label is located on the face of the cover, which is attached to the side of the HNU Posterior Component.
FRU	Field Replaceable Unit part number
FIELD	Field Strength
REF REV.	Revision of the part
	The electronic Instructions For Use (eIFU) symbol means that the operator manual can be viewed from a PC.
	P-connector locked symbol. Connector is locked in place and ready for scanning.
	P-connector unlocked symbol. Connector is not locked in place. Do not scan until the connector is locked.
	1.5T designates that the coil must be used on a 1.5T MR system
	3.0T designates that the coil must be used on a 3.0T MR system

Table 1-9: Symbol and Label Definition (Continued)

Symbol	Definition
	Indicates that the connector is to be plugged into port A.
	Indicates that the connector is to be plugged into port B.
	Symbol indicates that this product is a medical device
	P-port symbol that represents the P-port that you should plug the connector into on the MR system. The “X” will be replaced on the actual label with the numbers 1, 2, 3, or 4, which will correspond with one of the P-ports on your system.
	P-port symbol with a solid slash represents a P-port that you should not plug the connector into on the MR system. The coil will not work if plugged into this P-port. The “X” will be replaced on the actual label with the numbers 1, 2, 3, or 4, which will correspond with one of the P-ports on your system.
	P-port symbol with a broken slash represents a P-port that you may be able to plug the connector into on the MR system. Depending on your particular MR system configuration, the coil may work if plugged into this P-port. The “X” will be replaced on the actual label with the numbers 1, 2, 3, or 4, which will correspond with one of the P-ports on your system.
	European Representative

Table 1-9: Symbol and Label Definition (Continued)

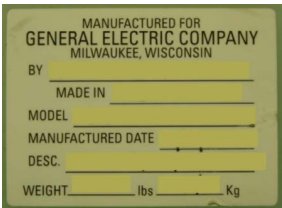
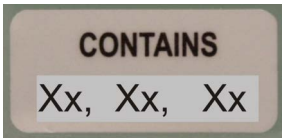
Symbol	Definition
	Swiss Authorized Representative
	Radio Frequency Radiation symbol
	Phantom Identifier
	Phantom Contents
	Phantom Positioning
	CE label
	Fragile - Do not drop or handle package roughly

Table 1-9: Symbol and Label Definition (Continued)

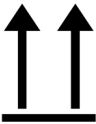
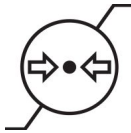
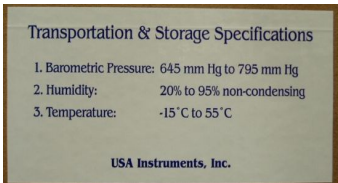
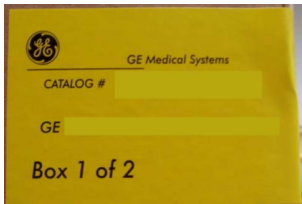
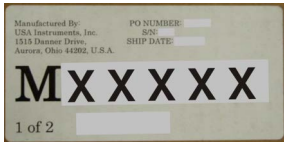
Symbol	Definition
	Keep the packaging out of the rain. Keep packaging dry.
	This end up. The arrows point to the side of the package that is to remain facing up.
	Legacy Do not allow package to freeze
	Legacy Do not drop or handle package roughly
	Legacy Label. Keep packaging dry. Do not drop or handle package roughly. The arrow points to the side of the package that is to remain facing up.
	Temperature Limitation

Table 1-9: Symbol and Label Definition (Continued)

Symbol	Definition
	Humidity Limitation
	Atmospheric Pressure Limitation
	Legacy Transportation and Storage Specifications
	Shipping Content label. Identifies the catalog number, GE part number and the number of boxes in the shipment.
	Identifies contents, purchase order number, part number, quantity, shipping slip number, serial number and date shipped.
	Shipping label. Identifies manufacturer, purchase order number, serial number, ship date and part numbers and number of boxes in the shipment.

Coil Description

The 1.5T HD Head, Neck, and Spine Array and the 3.0T Head, Neck, and Spine Array Coils are receive-only 16-channel, 29-element coils designed for use with GE 1.5T MRI Systems and GE 3.0T MRI Systems, respectively. These coils incorporate soft, flexible components that conform to patients' anatomy, accommodating various body contours while minimizing patient discomfort.

Throughout this manual the 1.5T HD Head, Neck, and Spine Array and the 3.0T Head, Neck, and Spine Array coils will be referred to individually as the 1.5T HNS Coil and the 3.0T HNS Coil or collectively as the 1.5T and 3.0T HNS Coils.

The 1.5T and 3.0T HNS Coils are comprised of 4 units: the Head/ Neck Unit (HNU), which includes the Anterior (face) unit, the Thoracic/ Lumbar unit (TLU), the Neck Chest Unit (NCU) and Adapter Block (Horseshoe).

The TL Unit, Horseshoe and NCU are identical on systems equipped with HD and P connectors. However, the HNU differs from system to system in the follow ways:

For the 1.5T HNU

- The HNU with Legacy Bendix connector and HD connector has two 8-channel cables.
- The HNU with one HD connector has one 16-channel cable.
- The HNU with one P connector has one 16-channel cable.

For the 3.0T HNU

- The HNU with 3.0T Legacy connector and HD connector has two 8-channel cables.
- The HNU with one HD connector has one 16-channel cable.
- The HNU with one P connector has one 16-channel cable.

NOTE: The HNS Coil is available for the software release version 14.0 and greater for the 1.5T coil and 15.x and greater for the 3.0T coil. It is important to make certain that your system coil connector is compatible with the connector on your coil. It is also important that you use the mode configuration names appropriate for the software you are running, otherwise faults may occur.

The 1.5T HD Head, Neck, and Spine Array and the 3.0T Head, Neck, and Spine Array Coils are designed to be used by a Registered MRI Technologist. The Registered MRI Technologist will operate the scanner from the control room. If the patient is claustrophobic another MRI Technologist or clinical staff member may stay in the magnet room with the patient.

Clinical Benefits of MR Coils

The clinical benefit of an MR Coil is to help the healthcare professional provide accurate diagnostic information that enhances the diagnostic and treatment care pathways of the patient for a variety of diseases and conditions.

Coil Specifications

Table 1-10: Coil Specifications

Field Strength	Model	Device Number	Frequency
1.5T	HD Head Neck and Spine Array	2416329	63.86 MHz
3.0T	3.0T 16CH Head Neck Spine (HNS) Array	5341333	127.73 MHz

Coil Dimensions and Weight

NOTE: Store the coil in an area that exceeds the dimensions of the coil.

Dimensions

122 cm Long x 46 cm Wide x 37 cm High (48 in Long x 18 in Wide x 15 in High)

Weight

Head-Neck Array 24.25 lb [11 Kg]

Neck-Chest Array 3.3 lb [1.5 Kg]

TL Spine Array 12 lb [5.5 Kg]

Coil Installation and Service

Coil installation and configuration must be completed by a GE Service Representative. Please contact a GE Service Representative for detailed coil installation and configuration information. The GE Field Engineer shall consult the appropriate service manual or service method CD for installation instructions.

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Chapter 2

Safety

Patient safety and comfort must be your primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards with the potential to cause injuries.

Route cables through the center of the magnet bore. Place cables under the cushion whenever possible. Routing near the sides of the bore increases the likelihood of cable heating from induced currents.

Keep the length of the cable in the bore to a minimum. When possible, avoid bending the cable 180 degrees. Route cables out of the bore in the most direct way possible, without looping or coiling.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. If the patient reports heating, burning or tingling sensations, **stop the scan immediately.**

All personnel using this coil must be instructed in its proper connection and handling. Personnel must observe all warnings and cautions appearing in this Manual.

Intended use for the 1.5T HNS Coil

The 1.5T Brain / Spine Array Coil is designed for use with the 1.5T MRI system manufactured by GE Healthcare. The coil is for imaging of the brain, cervical spine, soft tissues and vasculature of the head, neck, upper chest, thoracic and lumbar spine. The nucleus excited is hydrogen. The 1.5T Brain / Spine Array Coil is designed for use with the 1.5T MRI systems manufactured by GE Medical Systems.

Intended use for the 3.0T HNS Coil

The GE 3.0T 16 Channel Head Neck Spine Coil is a receive-only RF coil designed for use with the 3.0T MRI Systems manufactured by GE Healthcare. The coil is intended for imaging of the brain; cervical, thoracic and lumbar spines; as well as the soft tissues and vasculature of the head, neck and upper chest. The nucleus excited is hydrogen.

Contraindications

You should be aware of the contraindications for use related to the strong magnetic field of the MR system. Read this Safety chapter in this manual and then either the MR Safety Guide or the safety chapter in the operator manual provided with your system for contraindications. Follow screening instructions outlined in your MRI system documentation as well as the procedures established by your site. Use of this coil does not require additional MRI safety screening.

Patient Safety



WARNING:

Patient burn risk. Do not cross or loop coil cables. Do not loop RF receive coil cables and ECG leads.



WARNING:

Do not allow the coil cable to come into contact with the skin of the patient. Use a thermal resistant material or pad to keep the cable from touching the patient. Failure to do so may result in patient burns.



WARNING:

RF can cause localized heating at contact points between adjacent body parts when a loop is formed. Such localized heating can result in a tingling sensation, discomfort, or burns. Loops may be formed by clasped hands, crossed arms, legs or feet, when a patient's hands are touching another body part, or when a female patient's breasts are compressed to her chest. Use pads between body parts to avoid creating a loop with adjacent body parts.



WARNING:

Do not use accessories (e.g. pads or straps) that have not been specifically tested and approved for use in the MR environment. Use only GE recommended accessories. Use of non-approved accessories may result in patient burns or injuries or image degradation. Even auxiliary devices labeled as compatible with MR equipment are capable of causing injury if the manufacturer's instructions are not followed.

**WARNING:**

RF can cause localized heating at contact points between the patient and the bore resulting in discomfort or burns. Place appropriate nonconductive padding between the patient and the bore wherever a portion of the body may come into contact with the magnet opening.

**WARNING:**

Always place appropriate non-conductive padding between the surface coil and the skin of the patient to prevent burn injuries.

**WARNING:**

Prior to patient placement in the coil, assure that any breached or compromised patient skin surfaces that come in contact with the coil have been adequately bandaged or covered.

**CAUTION:**

To avoid injuring the patient or damaging the coil, watch carefully for pinch points as the table moves into the bore. Stop advancing the table if the patient or any part of the coil comes into contact with the bore.

Equipment Safety



WARNING:

Auxiliary equipment (e.g. gating equipment, vital signs monitoring systems and RF coils) that has not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. Such equipment may also interfere with proper operation of this coil. Only use equipment approved for the MR environment.



WARNING:

This coil is not suitable for use in the presence of a flammable anesthetic mixture with air or with oxygen or nitrous oxide.



WARNING:

Patient burns may result from RF coupling of this device with other devices remaining in the magnet. Remove any unused accessory device, coil, or cable from the magnet before using this coil.



WARNING:

Route cables through the center of the magnet bore. Place cables under the pads when possible. Routing cables near the sides of the bore increases the likelihood of cable heating (from induced currents).



CAUTION:

Looped cables may cause RF coupling and degrade the scan performance of the coil. Do not cross or loop coil cables.



Electrical and Mechanical Safety



WARNING:

Electric shock hazard. This coil consists of electrical and mechanical components. Tampering with the coil by untrained personnel can be hazardous to the patients and equipment. Only properly trained and qualified personnel should service the coil.



WARNING:

Use only coils, cables and accessories that are in good condition. Before using the coil, visually check each coil component, cable and accessory to ensure that there is no external damage. If any coil component, cable or accessory is suspected of not being in good condition, discontinue use and contact a GE Service Engineer.



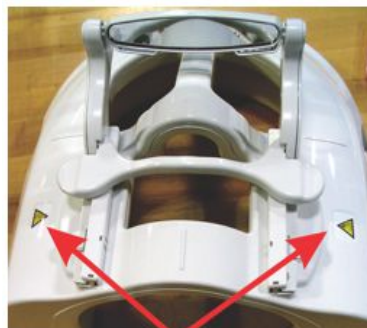
WARNING:

Electric shock hazard. No user serviceable parts. Refer service to qualified service personnel.

Definitions for Labels Located on the Coil

**WARNING:**

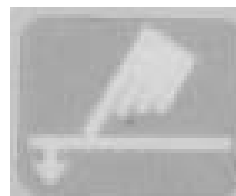
The arrows in the image to the right point towards two pinch hazard labels located on the HNU anterior unit. Do not allow the patient to place their hands on the anterior unit of the coil when the table is moving into the bore. The hands or arms of the patient may be pinched between the coil and the bore.

**WARNING:**

The arrow in the image to the right points toward a surface that may be hot, which is located on the HNU Posterior component. When scanning patients *without* the chest component, always place the connector cover securely over the connector on the HNU Posterior component. The cover prevents the hands of the patient from coming into contact with the connector. Failure to comply will leave the connector exposed, placing the patient at risk for RF burns or shock if the patient touches the connector during a scan.

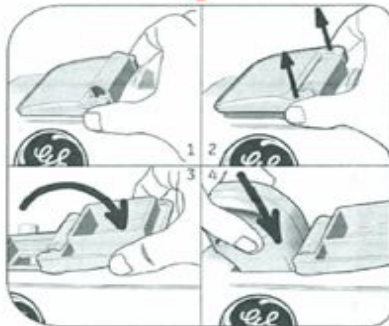
**CAUTION:**

This label, affixed to the Anterior Face, Horseshoe and Chest connector informs the user to firmly press down on all coil components to ensure that they are fully engaged and connected. Failure to do so may cause artifacts and decrease image quality.





The arrow in the image shown to the left points towards the label that shows how to properly lift and pivot the NCU connector cover away from the connector. Window number 4 then shows the NCU connector being plugged into the connector on the HNU Posterior component.



This label is attached to the NCU connector cover, which is located on the HNU Posterior component.



The label shown to the left can be found in numerous places down the sides of the HNU and TLU coil components. The “X” in the image to the left will be replaced by the actual number of the element.

For non-P-connector HNS Coils: The labels shown to the left may be found on the coil connector(s). The labels found on your coil are dependent on the MR system that the coil is to be used on, so the coil you have will not have all of the labels shown.



A Label. This label may be found on the top of your coil connector. It indicates that the coil is to be plugged into port A.



B label. This label may be found on the top of your coil connector. It indicates that the connector is to be plugged into port B.



The label to the left may be found on your coil connector. It indicates to not insert the connector with this side up. If connector is inserted with this side up, the coil will not operate.

For HNS coils with P-connectors, the labels to the right may be found on the P-connectors used on the PET/MR Coil Suite components.

- **The upper portion of the P-port connector label may contain a 1.5T or a 3.0T (location designated by X.XT in the photographs to the right). A 1.5T indicates that the coil must be used on a 1.5T MR system, and a 3.0T designates that the coil must be used on a 3.0T MR system.**
- **Locked and Unlocked symbols. Plug the P-connector into the system P-port with the red unlocked symbol showing. Then, turn the gray spindle handle clockwise until the green locked symbol is showing. The coil is ready for scanning when the green locked symbol is visible.**
- **The “D” shaped symbols with numbers 1, 2, 3, or 4 designate the P-connectors on the system.**
 - The P-port with no slash indicates that you should plug the connector into this P-port on the MR system.
 - The P-port symbol with a solid slash indicates that you should not plug the connector into this P-port on the MR system. The coil will not work if plugged into this P-port.
 - The P-port symbol with a broken slash indicates that you may be able to plug the connector into this P-port on the MR system. Depending on your particular MR system configuration, the coil may work if plugged into this P-port.



The serial number label for the coil also contains several symbols. This label specifies date of manufacture, place of manufacture, part number and revision. This label also has several symbols that are defined in this manual that will help the user better understand the specifics of the coil.

Modifications

Any changes or modifications to this coil must be approved and performed by GE prior to installation.

Emergency Procedures and Incident Reporting

If a Patient Requires Emergency Assistance

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.
- Please refer to the MR Safety chapter in your MR system operator manual or refer to the MR Safety Guide, part number 2381696, if you received one with your system, for patient emergency procedures.

Reporting of Serious Incidents

Any serious incident related to the use of this GE Healthcare device should be reported to both the manufacturer and the health authority/competent authority where the device is installed.

To report to GE Healthcare:

- Either contact your local service representative
- Or report to: In-box.complaints@gehealthcare.com

Please provide the following information:

- The catalogue number or the model designation of the device as stated on its identification plate affixed on the device
- The SystemID/serial number/lot number of the device
- Date of incident
- Description of incident, including any patient or user impact/injury
- Your contact information (facility, address, contact name, title, and telephone number)

Reporting of Incidents with the Coil

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise:

- Contact your local service representative to report the incident.
- Do not use the coil until a GE Service Representative inspects the coil and indicates it is safe to use.

Users should immediately report to GE any incidents or injuries that result from coil operation involving a patient, operator, or maintenance employee.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation has been conducted.



CAUTION:

Users should place a service call any time the coil is dropped or mishandled. A GE Service Representative should inspect the coil after it has been dropped to ensure it is safe to use.

MR Safety Information

Please refer to MR Safety chapter in your MR system operator manual or refer to the MR Safety Guide, part number 2381696, if you received one with your system. The MR Safety documentation describes the safety information you and the physicians must understand thoroughly before you begin to use the MR system. If you need additional training, seek assistance from qualified GE personnel.

This equipment is intended for use by qualified personnel only.

This manual should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this manual before attempting to use this product.**

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Chapter 3

Using the HNS Coils

Coil Installation

Coil installation is to be completed by a GE Service Representative or a third-party service provider. To have the coil installed, please call your GE Service Representative or your third-party service provider. Service representatives have access to documents that contain instructions for safely and compliantly installing, testing, and maintaining your MR Coil.

If you do not have a GE Service Representative or third-party service provider, use the Class A Service CD that was shipped with your MR System. This CD contains information on phantom setup and the usage of the MCQA tool. Please contact GE with any questions or concerns.

Lifting and Transporting the Coil Components

Pick up the individual coil components, HNU, Anterior, Adapter Block, NCU and TLU, by firmly gripping the components on both sides.

To carry the HNU when the anterior is on the posterior, firmly grip and carry the HNU anterior/posterior combination with hands on both sides and underneath the HNU posterior.



Do not carry any of the coil components by the cable. Damage to the coil components may occur. The coil may not work if damaged.

CAUTION:



Do not pick up the HNU posterior by the Anterior when the Anterior is attached to the posterior. Coil components may be damaged if dropped.

CAUTION:



Do not pick up the Anterior or HNU posterior and Anterior combination by the mirror. Coil components may be damaged if dropped.

CAUTION:

Installing the mirror onto the HNU Anterior (Face) Unit

To prevent breakage, the mirror is designed to break away from the anterior unit. In the event that the mirror assembly is pulled from the anterior unit, the mirror can be reinstalled using the following procedure:

NOTE: If the mirror assembly has come apart or the snap features on the top rails are broken, contact your GE service representative for replacement. A top mirror that gets pulled out of a socket is an example of a failure that may occur. See figure 3-1 for an example. If this occurs, please contact your GE service representative for replacement or repair.

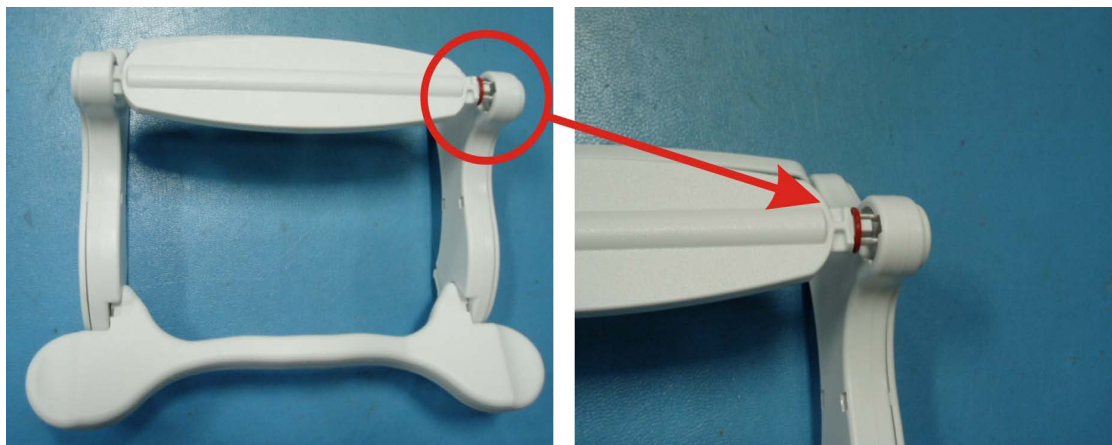


Figure 3-1: Example of the top mirror pulled out from the socket

1. If the mirror has come off of the anterior unit and the top rails (shown in figure 3-1) are still attached to the mirror, push the locking bar on the mirror into the up position (shown in figure 3-9). This will release the top rails from the mirror so that the mirror can be re-installed onto the anterior unit as described in the following steps.
2. Attach the top rails (shown in figure 3-2) onto the bottom rails that are screwed onto the anterior unit (figure 3-3). The arrows in figure 3-3 point towards the rails.

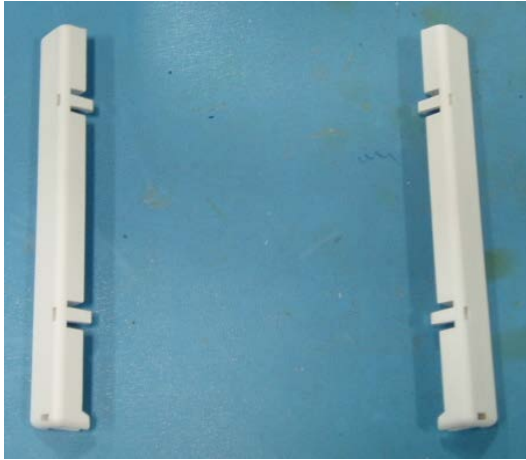


Figure 3-2: Top Rails



Figure 3-3: Arrows point toward the bottom rails that are attached to the anterior unit

NOTE: Steps 3 and 4 instruct how to install the right top rail onto the right bottom rail. To attach the left top rail, simply follow the same procedure, but use the pieces for the left side. The part numbers for the top rails can be helpful to distinguish the two top rails apart. The part number for the right top rail is 2419526, and the part number for the left top rail is 2419525. Figure 3-4 shows the location of the part numbers on the rails.

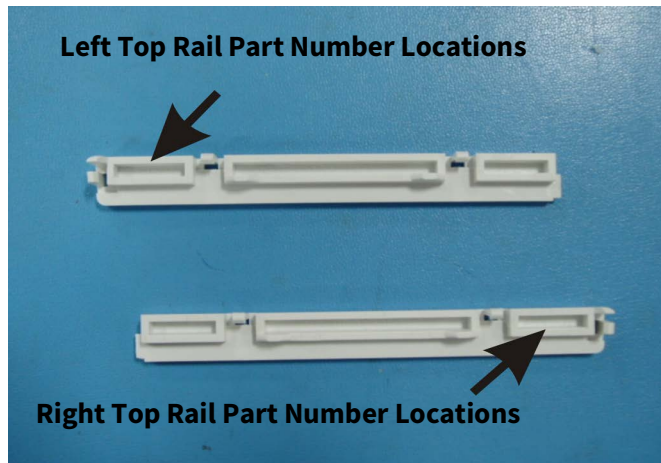


Figure 3-4: Top rail part number locations

3. Insert the front tab on the top right rail into the front cut-out of the bottom rail (see figure 3-5 and 3-6).

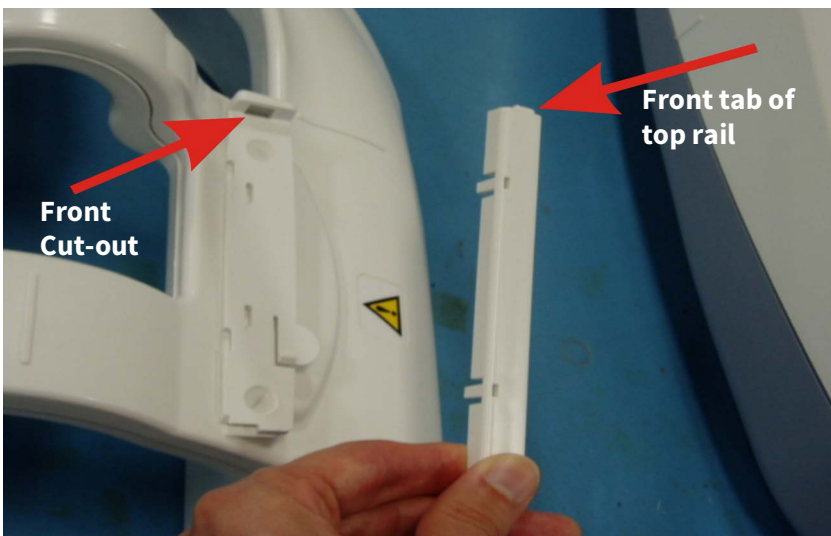


Figure 3-5: The front tab of the top rail and the front cut-out of the bottom rail

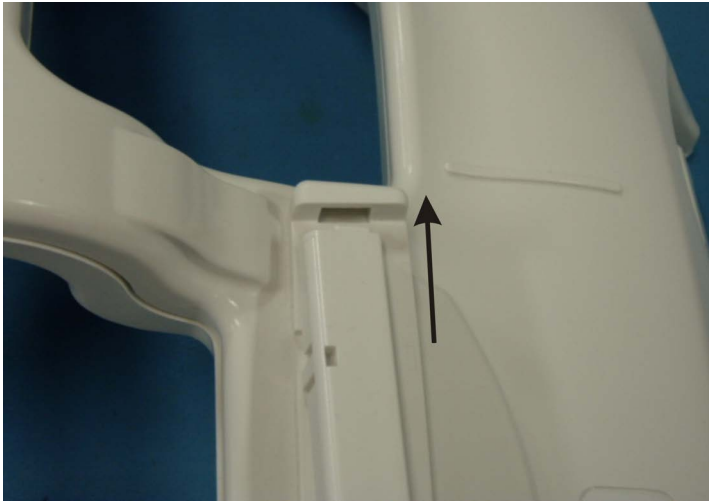


Figure 3-6: Inserting the top rail tab into the bottom rail

4. Gently push the top rail down onto the bottom rail (figure 3-7). A click will be heard when the top rail is secured to the bottom rail. Figure 3-8 shows the top rail on the bottom rail.



Figure 3-7: Gently push the top rail onto the bottom rail



Figure 3-8: Top rail on the bottom rail

5. If needed, attach the opposite rail using the same method as described in steps 3 and 4.
6. Push the locking bar of the mirror into the up position. See figure 3-9.

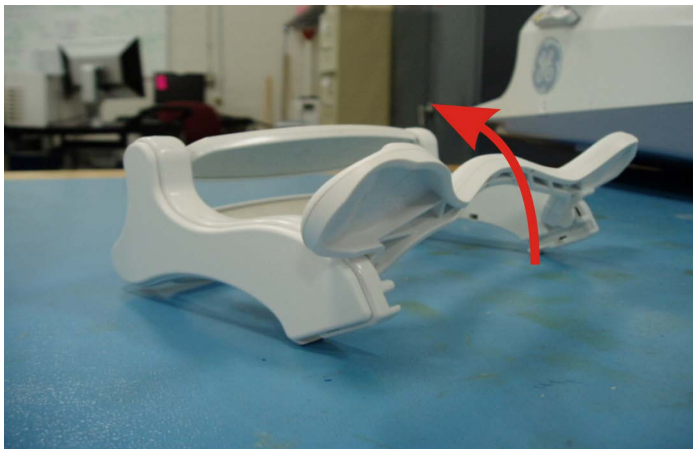


Figure 3-9: Locking bar in the up position

7. Align the channels at the bottom of the left and right side of the mirror with the channel of the left and right rails. See figure 3-10.



Figure 3-10: Mirror channel aligned with the rail channels

8. Slide the mirror onto the rails as shown in figure 3-11. The mirror needs to slide past the rail tabs. See figure 3-12 for a photograph of a rail tab.

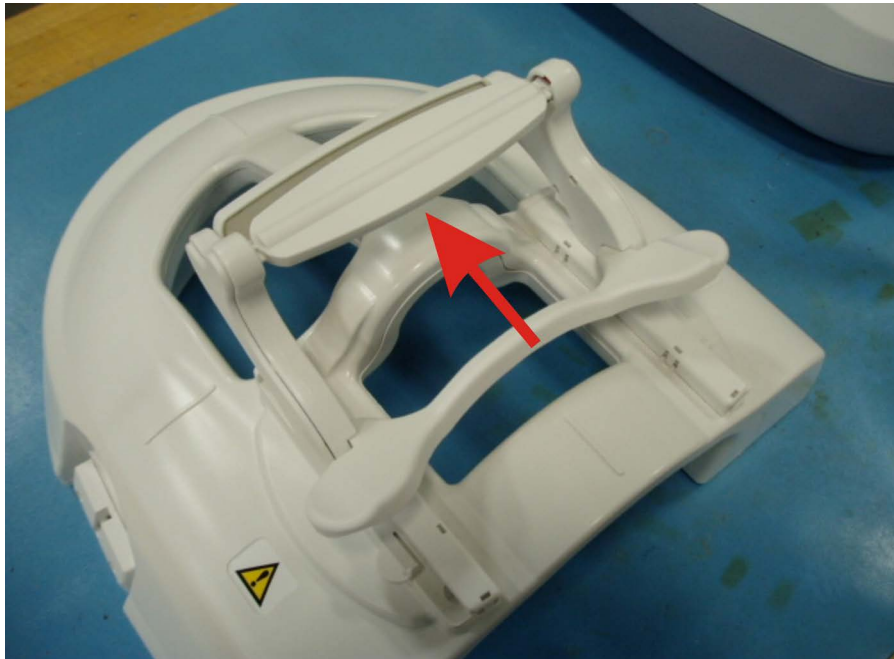


Figure 3-11: Slide mirror onto the rails

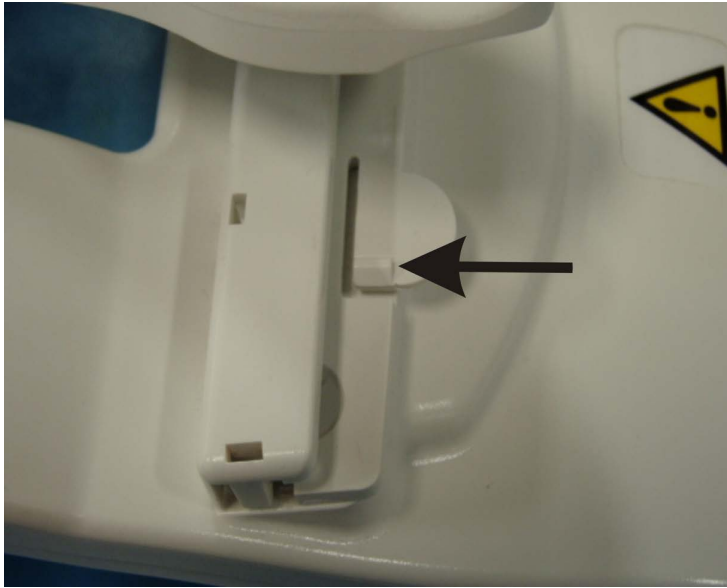


Figure 3-12: Right side rail tab.

9. Gently push the locking bar on the mirror down to lock the mirror in place. See figure 3-13.



Figure 3-13: Push the locking bar down

10. The mirror is now installed onto the anterior unit.

Coil Components

The 1.5T and 3.0T HNS Coils are composed of five separate components. The element groups are marked 1-6 on the sides of the coil for positioning and landmarking purposes (please refer to Figure 3-14 below):

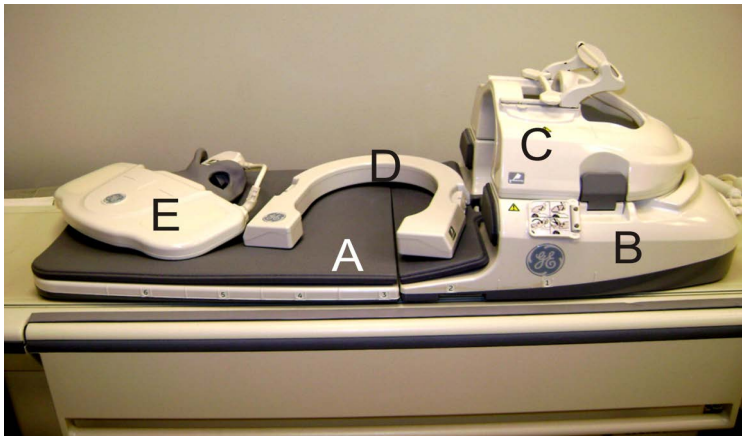


Figure 3-14: Components of the HNS (the groups refer to the element distribution of the coil. See Figure 3-15 for element distribution):

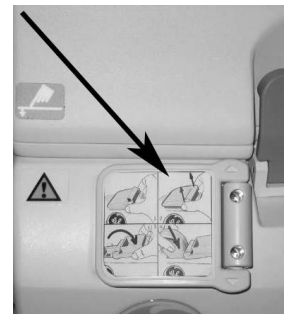
- A = TL Spine (TL) (groups 3 through 6)
- B = HNU Posterior Unit (Posterior Head and groups 1 and 2)
- C = HNU Anterior Unit (Face)
- D = Anterior Adapter Block (Horseshoe)
- E = Neck Chest Unit (Chest)

The components may be used all together, or in several combinations. The TL component may be used independently. Please refer to Table 3-1 and Table 3-4 for more information on components and configurations.



WARNING:

When scanning patients *without* the chest component, always place the connector cover securely over the connector on the Posterior Head Component. The cover prevents the patient's hand from coming in contact with the connector. Failure to comply will leave the connector exposed, placing the patient at risk for RF burns or shock if the patient touches the connector during a scan.





CAUTION:

The Revision 3 1.5T HNS Coil is not compatible with HNS Coil Revision 2 components, with the exception of the TL Spine, which is interchangeable. Connection between incompatible parts may be physically possible, but faults will occur. Check the label on each component for the revision number.



CAUTION:

1.5T HNS and 3.0T HNS coil components are not compatible. Check the labels on each component to ensure common field strength. Connection between incompatible parts may be physically possible, but faults will occur.



CAUTION:

The 3.0T HNS coil components manufactured by GE Healthcare Coils and the 3.0T HNS coil components manufactured by Medrad are not compatible. Check the labels on each component for common manufacturer's name before use. Using 3.0T HNS Medrad components with 3.0T HNS GE Healthcare Coil components may degrade performance or safety.

Element Distribution

There are a total of 29 elements in the 1.5T and 3.0T HNS Coils. Fig. 3-15 depicts the locations of the groups of elements. Use this information along with Table 3-1 on the following page to determine the correct configuration for your study.

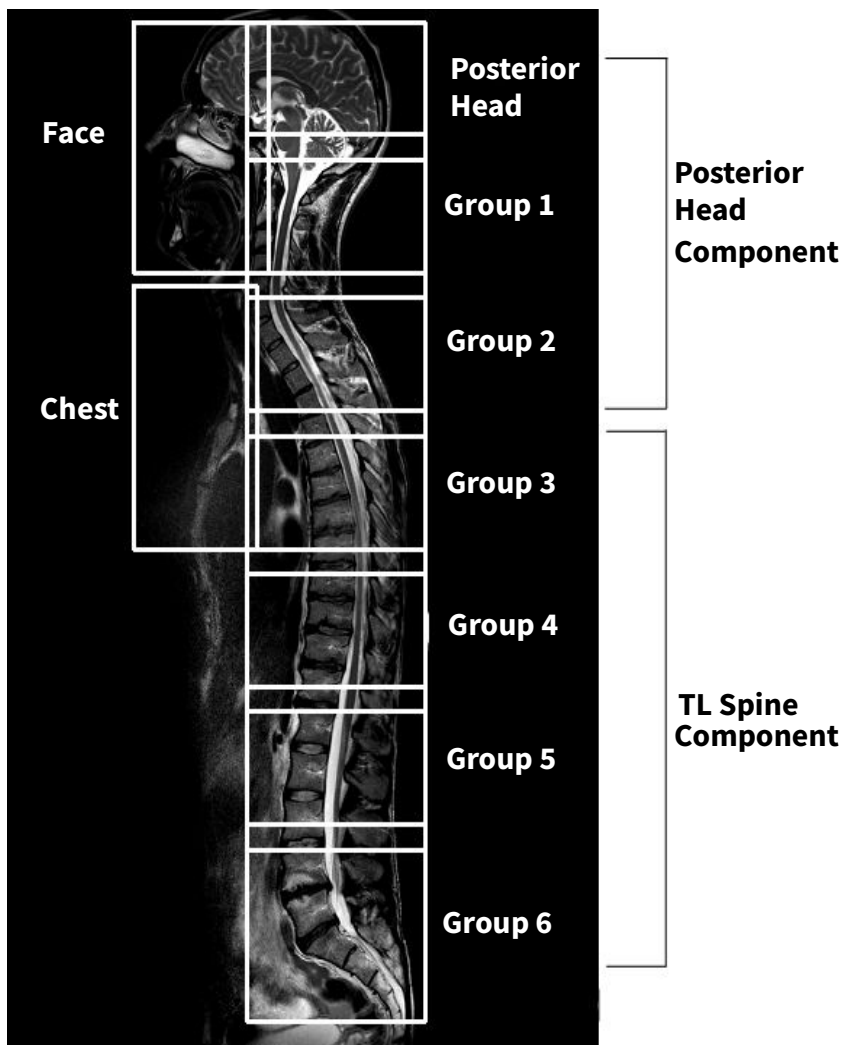


Figure 3-15: 1.5T and 3.0T HNS Coil element distribution

For Users of MR750, MR450, MR450w and Older Systems

The following sections, Coil Mode Configuration, Selecting the Correct Configuration, Using Configurations, and Using the Equivalent Spine Configurations, are for the MR750, MR450, MR450w and older systems.

Coil Mode Configuration

The 1.5T and 3.0T HNS Coils were designed to eliminate multiple coil usages per patient in order to increase throughput and patient comfort. These coils have 39 different configurations with which to scan. These configurations allow you to optimize field of view (FoV) when using a variety of coil components, and will enable you to scan the region of interest with excellent image quality without mandatory coil changes and patient disruption.

To optimize image quality, use the following section as a guide to help you select the proper components and coil configurations for your study.

The following Configuration Table explains the different coil components and configurations available to you. When the desired coil components are plugged into the system, the available configurations become selectable. Therefore, hardware (coil components) dictates the coil configurations that are available.

Select a configuration based on the anatomy of interest, and the FoV desired. Then center the anatomy to the element groups used in the desired configuration.

Table 3-1: Coil Configurations

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Horseshoe and TL Spine	HNS CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNS CTLMID	48cm	Largest FoV T and L spine	2,3,4 and 5	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNS CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL 34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No
Posterior Head and Horseshoe	HNS CTL12	28cm	Smallest FoV C spine	1 and 2	No

Table 3-1: Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Face and TL spine	HNS HEAD	30cm	Brain	Post Head, Face and 1	Yes
	HNS 35FOV	48cm	Brain, C spine	Post Head, Face and 1,2 and 3	No
	HNS 48FOV	48cm	Brain, C spine, T spine	Post Head, Face and 1,2,3 and 4	No
	HNSp CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNSp CTLMID	48cm	Largest FoV T and L spine	2,3,4 and 5	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNSp CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNSp CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNSp CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNSp CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No
Posterior Head and Face	HNSp CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS HEAD	30cm	Brain	Post Head, Face and 1	Yes

Table 3-1: Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Face and Chest	HNS Neck	30cm	Smallest FoV Carotids, C spine, ST Neck	Face, 1,2 and 2 Chest elements	Yes
	HNS NeckChest	35cm	Large FoV Carotids, C spine, ST Neck	Face, 1, 2 and all 5 Chest elements	Yes
	HNS NVFull	48cm	Largest FoV Carotids, C spine, ST Neck	Post Head, Face,1,2 and all 5 Chest Elements	Yes
	HNS NVHead	30cm	Brain	Post Head, Face and 1	Yes
	HNS-CTL12	28cm	Smallest FoV C spine	1 and 2	No

Table 3-1: Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Face, TL Spine and Chest	HNS Neck	30cm	Smallest FoV Carotids, C spine, ST Neck	Face,1,2 and 2 Chest elements	Yes
	HNS NeckChest	35cm	Large FoV Carotids, C spine, ST Neck	Face,1,2 and all 5 Chest elements	Yes
	HNS NVFull	48cm	Largest FoV Carotids, C spine, ST Neck	Post Head, Face,1,2 and all 5 Chest elements	Yes
	HNS NVHead	30cm	Brain	Post Head,Face and 1	Yes
	HNS BrachPlx	40cm	Brachial plexus, C spine	1,2,3 and all 5 Chest elements	Yes
	HNS-CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNS-CTLMID	48cm	Largest FoV T and L spine	2,3,4 and 5	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS-CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNS-CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS-CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS-CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No

Table 3-1: Coil Configurations (Continued)

Components Connected	Configurations Available	Max. FoV	Best Use	Element Groups Used	Asset
Posterior Head, Horseshoe, TL Spine and Chest	HNS BrachPlex	40cm	Brachial plexus, C spine	1,2,3 and all 5 Chest elements	Yes
	HNS_CTLTOP	48cm	Largest FoV C and T spine	1,2,3 and 4	No
	HNS_CTLMID	48cm	Largest FoV T and L spine	1,2,3 and 4	No
	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS_CTL123	40cm	Large FoV C and T spine	1,2 and 3	No
	HNS_CTL234	40cm	Large FoV T spine	2,3 and 4	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS_CTL12	28cm	Smallest FoV C spine	1 and 2	No
	HNS_CTL23	28cm	Smallest FoV Upper T spine	2 and 3	No
	HNS CTL34	28cm	Smallest FoV Mid/Low T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV Lower T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No
TL only	HNS CTLBOT	48cm	Largest FoV T and L spine	3,4,5 and 6	No
	HNS CTL345	40cm	Large FoV T and L spine	3,4 and 5	No
	HNS CTL456	40cm	Large FoV T and L spine	4,5 and 6	No
	HNS CTL34	28cm	Smallest FoV Lower T spine	3 and 4	No
	HNS CTL45	28cm	Smallest FoV T/L spine	4 and 5	No
	HNS CTL56	28cm	Smallest FoV L spine	5 and 6	No

Selecting the Correct Configuration

Coil configuration nomenclature is based on the coil components currently connected to the MRI system. For example, with spine configurations, a dash in the configuration name indicates that, along with other components, the Chest component is currently connected to the MR system; an underscore in the name indicates that, along with other components, the Horseshoe **and** Chest components are connected to the system. See Table 3-2 for configuration file naming conventions.

Table 3-2: Configuration File Naming Conventions

Character	Configuration Name	Posterior Head	Face	TL Spine	Chest	Horseshoe
“p”	HNSp CTL12	Yes	Yes	Optional	No	No
“(–)” Dash	HNS-CTL12	Yes	Yes	Optional	Yes	No
“()” Underscore	HNS_CTL12	Yes	No	Optional	Yes	Yes

There are 39 different coil configurations available with the Head, Neck and Spine Coils. The correct selection for your study depends on the following:

- Components connected
- Type of examination (multiple exams on the same patient)
- Desired FoV
- Patient acceptance (when claustrophobia/anxiety prevents using Face component)
- Image quality (if the desired image quality is not realized, an alternate configuration may be chosen)

NOTE: Selecting **[Currently Connected]** from the Coil Selection list narrows the list of usable coil configurations. GE recommends using the Currently Connected feature before selecting configurations.

Some configurations are only compatible when certain coil components are plugged into the system. If the configuration you want is not listed in the Coil Configuration screen, it may be necessary to add or remove coil components before your desired configuration is selectable.

Precise patient positioning and proper FoV will yield optimum results. Improper use of the coil is the major cause of image artifacts.

- Refer to Table 3-1 for configurations optimized for ASSET applications.
- PROPELLER is compatible only with HNS Head and HNS NVHead configurations.
- Gradient Coil Zoom mode is not recommended when using HNS CTL45 and HNS CTL456.

Using Configurations

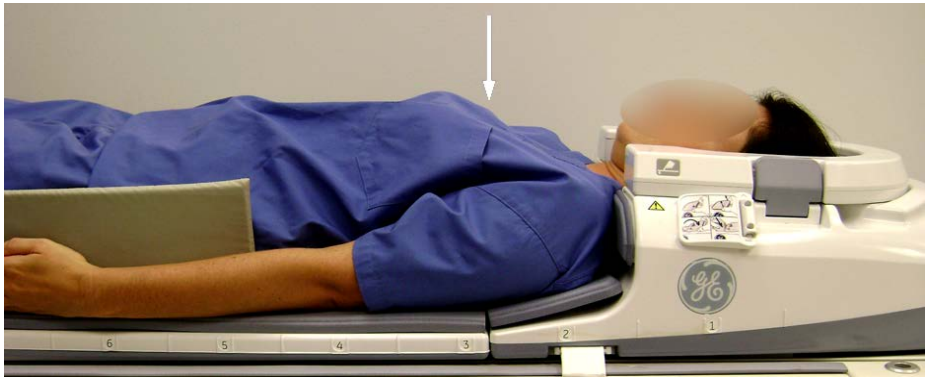
Please refer to the following section for detailed information concerning the 39 different HNS Coil configurations. The coil components are listed, along with the activated element groups, for each available configuration. Please refer back to Figure 3-15 for the Element Group name and number locations, if needed.

HNS CTLTOP

Components:	Posterior Head, Horseshoe, TL Spine
Element Groups:	1, 2, 3, and 4 (48cm FoV)

	Example Image 
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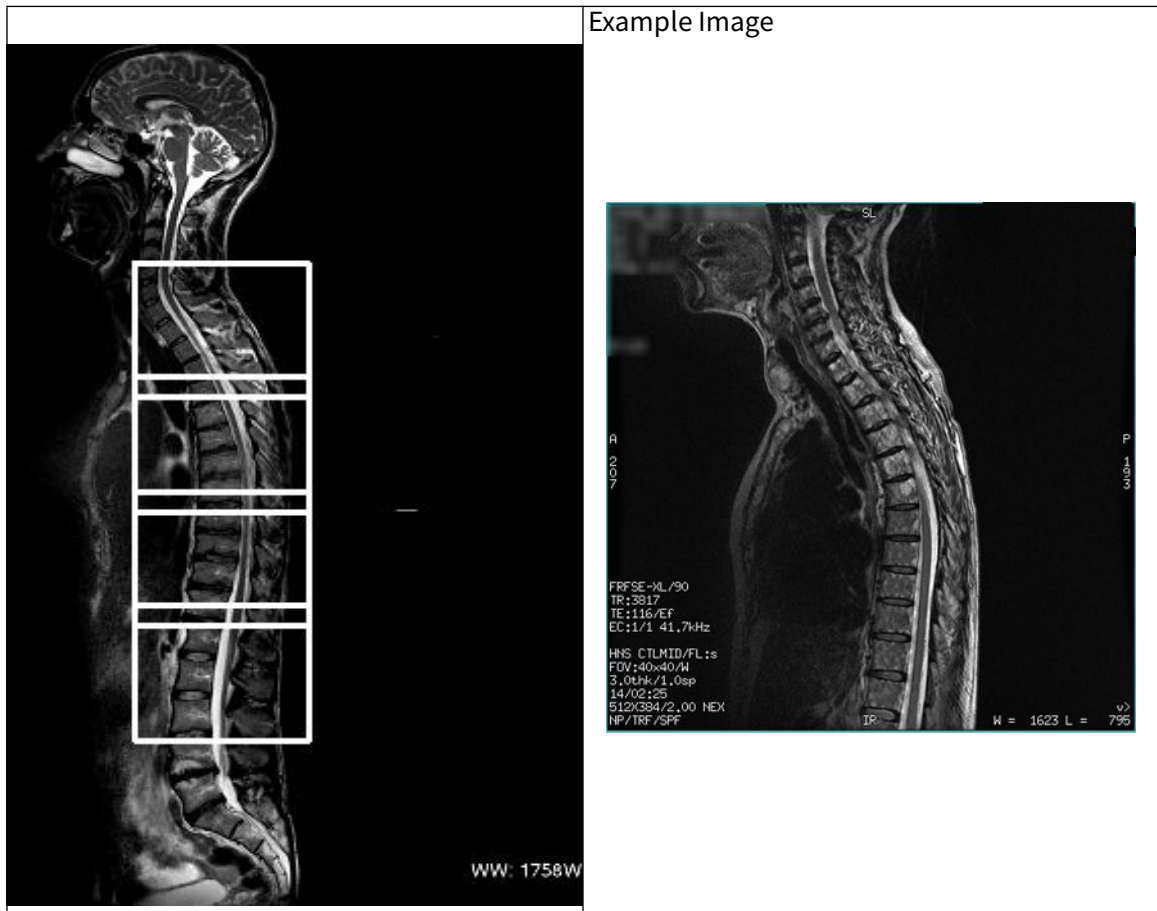
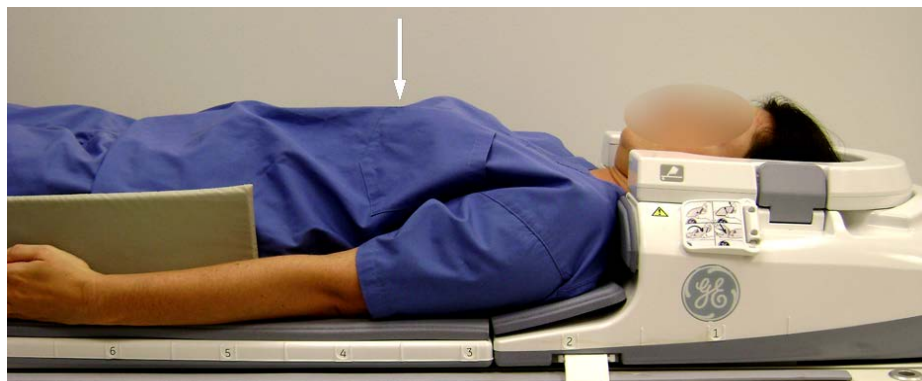
Landmarking:



Best Use:	Largest FoV Cervical and Thoracic spine.
-----------	--

HNS CTLMID

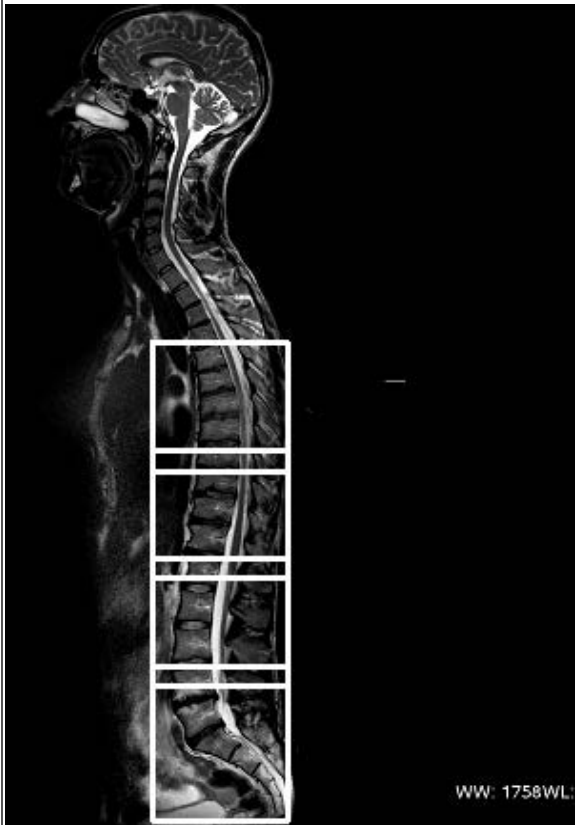
Components:	Posterior Head, Horseshoe, TL Spine
Element Groups:	2, 3, 4, and 5 (48cm FoV)

**Landmarking:**

Best Use:	Largest FoV Thoracic and Lumbar spine.
------------------	--

HNS CTLBOT

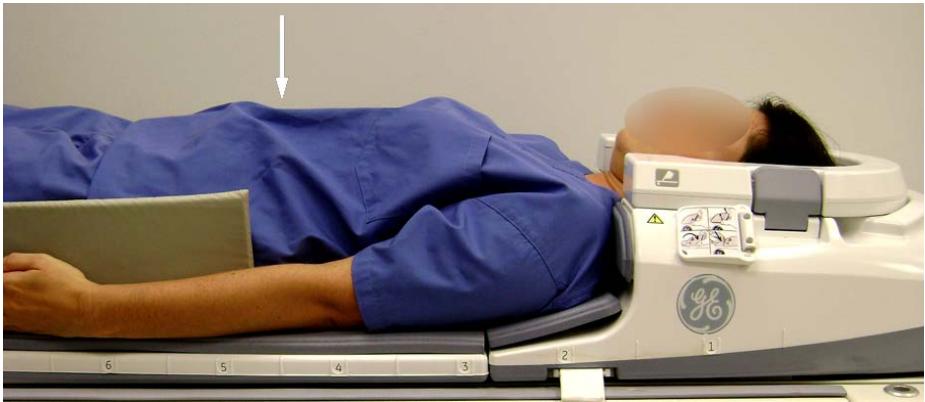
Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	3, 4, 5 and 6 (48cm FoV)



WW: 1758WL

Example Image

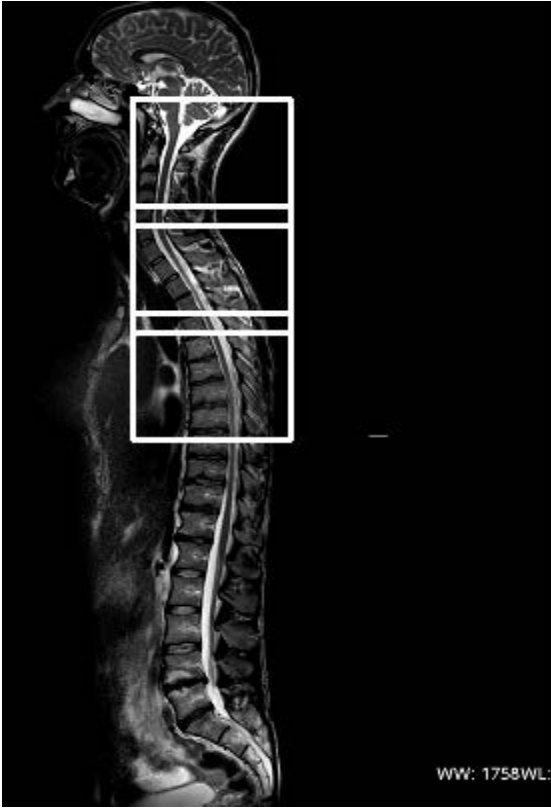
Landmarking:



Best Use:	Largest FoV Thoracic and Lumbar spine.
-----------	--

HNS CTL123

Components:	Posterior Head, Horseshoe, TL Spine
Element Groups:	1, 2 and 3 (40cm FoV)

 WW: 1758WL:	Example Image 
---	--

Landmarking:

Best Use:	Large FoV Cervical and Thoracic spine
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HNS CTL234

Components:	Posterior Head, Horseshoe and TL Spine
Element Groups:	2, 3 and 4 (40cm FoV)



WW: 1758WL

Example Image



T1FLAIR
TR:2359
TE:21.5/EF
EC:1/1 41.7kHz
TI:824
HNS CTL234/FL:s
FOV:35x35/W
4.0thk/1.0sp
7/02:39
612x256/2.00 NEX
St:1a/NP/SQ/TRF/SPF
T 273 W = 1126 L = 575

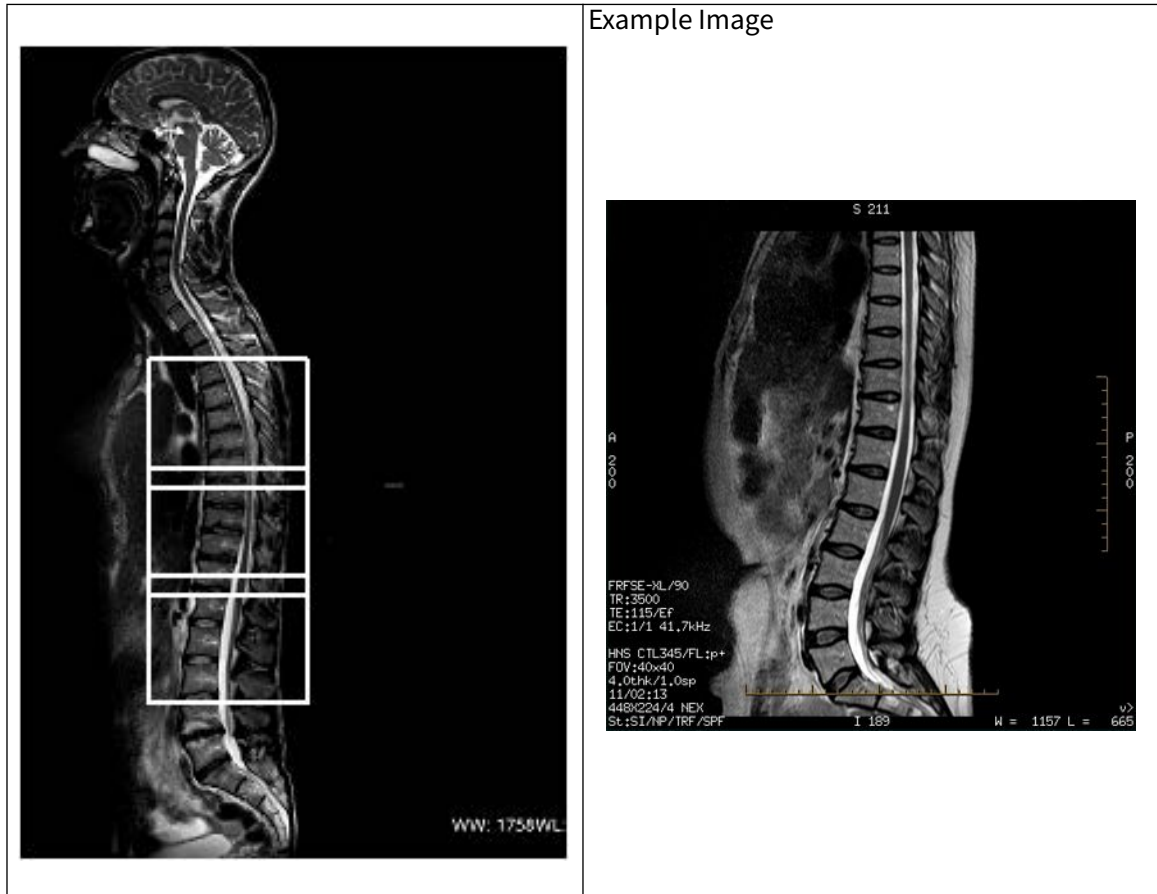
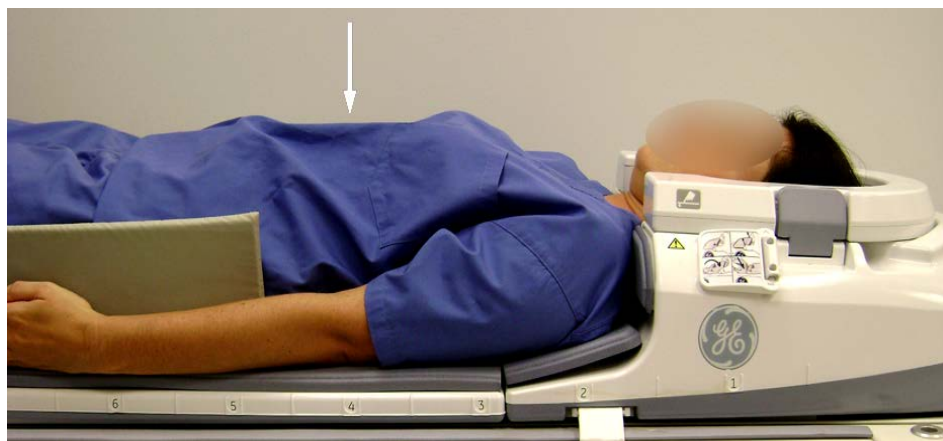
Landmarking:



Best Use:	Large FoV Thoracic spine.
-----------	---------------------------

HNS CTL345

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	3, 4 and 5 (40cm FoV)

**Landmarking:**

Best Use:	Large FoV Thoracic/Lumbar spine.
------------------	----------------------------------

HNS CTL456

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	4, 5 and 6 (40cm FoV)

	Example Image  T1FLAIR TR:2038 TE:27.3/Ef EE:1/1 31.2kHz TI:860 HNS CTL456/FL:p+ FOV:40x40 4.0thk/1.0sp 11/01:53 320x253/1.00 NEX NP/SD/TF/2512
--	--

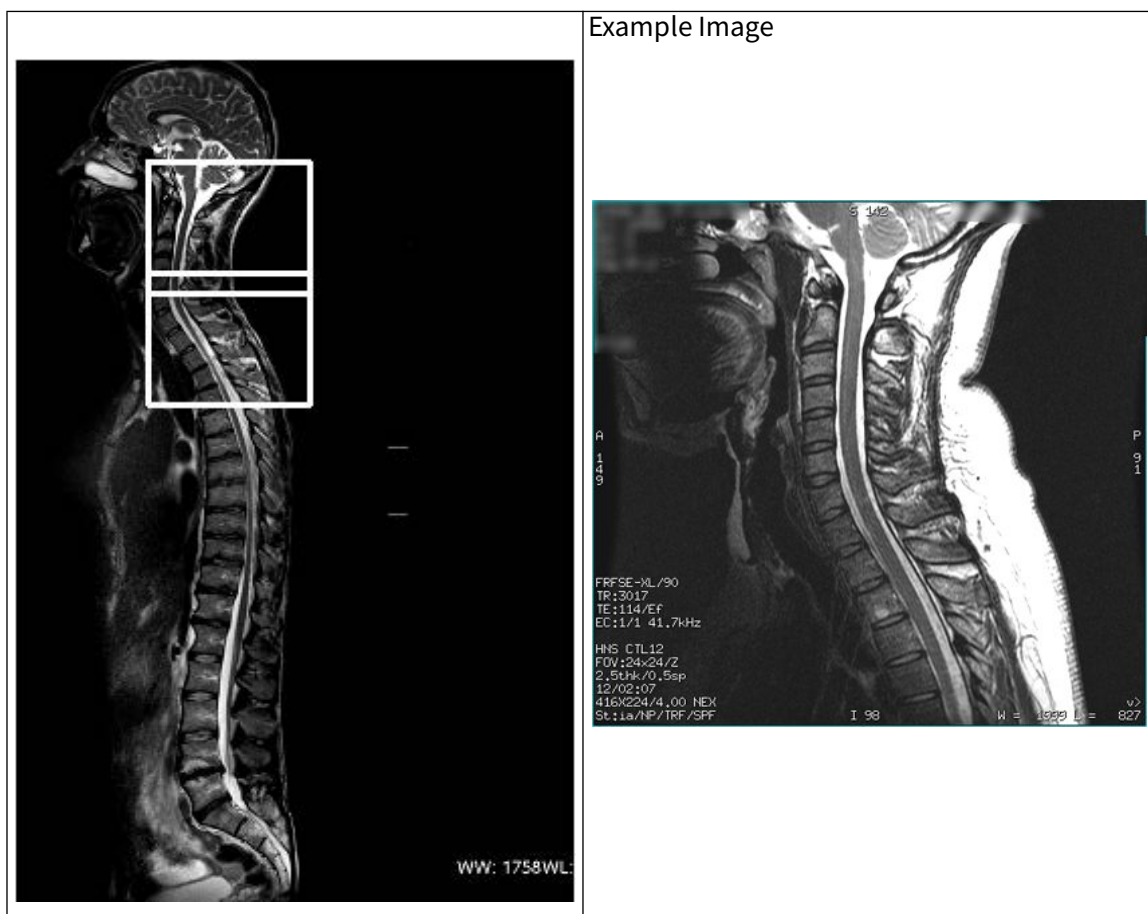
Landmarking:



Best Use:	Largest FoV Thoracic/Lumbar spine.
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HNS CTL12

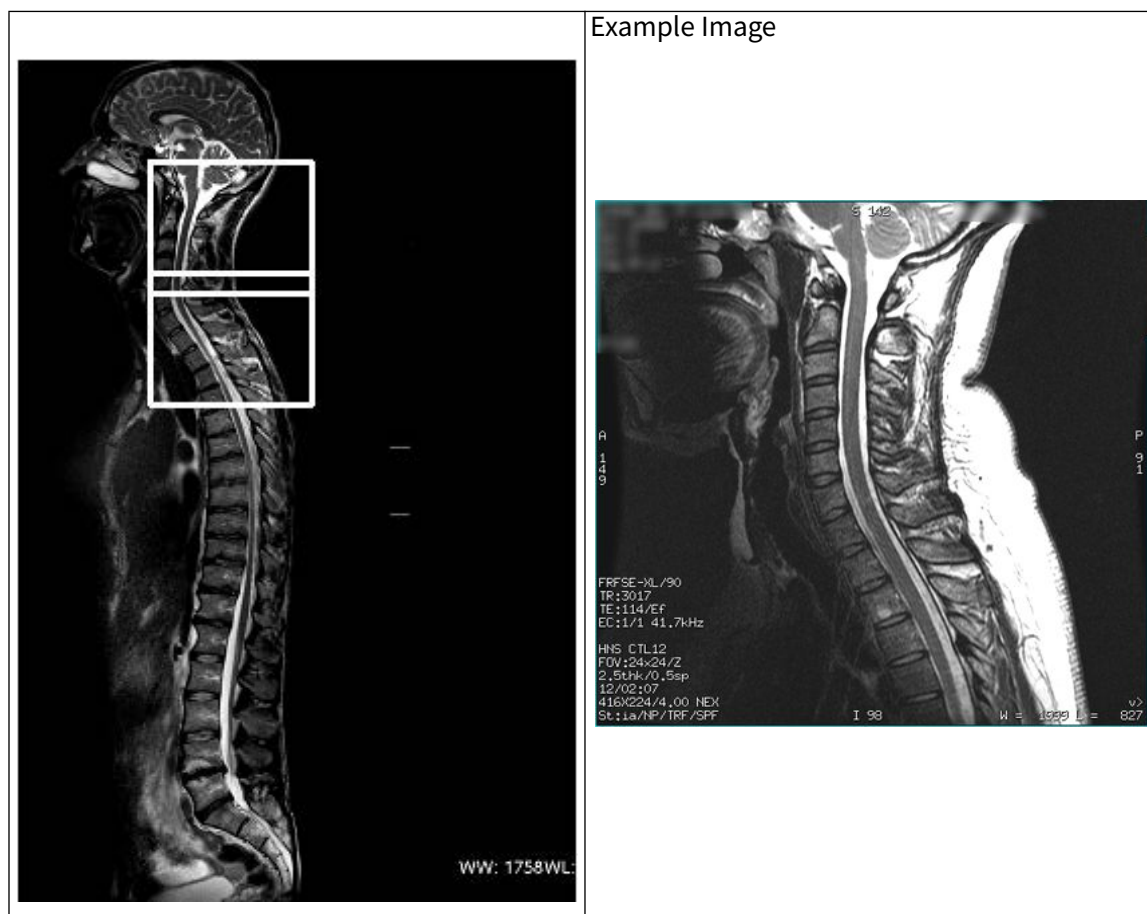
Components:	Posterior Head and Horseshoe (with or without TL Spine)
Element Groups:	1 and 2 (28cm FoV)

**Landmarking:**

Best Use:	Smallest FoV Cervical Spine
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HNSp CTL12

Components:	Posterior Head and Face (with or without TL Spine)
Element Groups:	1 and 2 (28cm FoV)



Landmarking:



Best Use:	Smallest FoV Cervical spine
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HNS-CTL12

Components:	Posterior Head, Face and Chest (with or without TL Spine)
Element Groups:	1 and 2 (28cm FoV)

 WW: 1758WL:	Example Image 
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Landmarking:

Best Use:	Smallest FoV Cervical spine
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HNS CTL23

Components:	Posterior Head, Horseshoe and TL Spine
Element Groups:	2 and 3 (28cm FoV)



WW: 1758WL:

Example Image



FRFSE-XL90
TR:3500
TE:106.6/ef
EC:1 / 51.2kHz

HNS CTL23
FOV:26x26AW
3.0MM/1.0sp
12/03/09
320x256/4.00 NEX
FC/NP/TRF/SPF

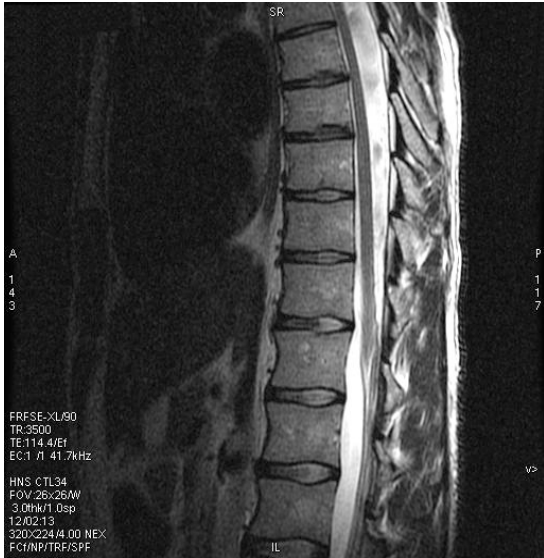
Landmarking:



Best Use:	Smallest FoV Upper Thoracic spine
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HNS CTL34

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	3 and 4 (28cm FoV)

	Example Image 
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Landmarking:

Best Use:	Smallest FoV Lower Thoracic spine
------------------	-----------------------------------

HNS CTL45

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	4 and 5 (28cm FoV)

	Example Image  FFSE-ML/90 TR:4300 TE:105/Ef EC:1/1 41.7Hz HNS CTL45/FL:s FOV:26x26/2 3.0thk/1.0sp 12/03:31 416x288/4.00 NEX NP/TRF/SFF I 103 W 2089 L 1184 WW: 1758WL
--	---

Landmarking:



Best Use:	Smallest FoV Thoracic/Lumbar spine
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HNS CTL56

Components:	TL Spine (with or without Posterior Head and Horseshoe)
Element Groups:	5 and 6 (28cm FoV)

**Landmarking:**

Best Use:	Smallest FoV Lumbar spine.
------------------	----------------------------

HNS Head

Components:	Posterior Head and Face (with or without TL)
Element Groups:	Posterior Head, Face, and 1 (30cm FoV)

	Example Image  FSE/Prop TR:4650 TE:126/Ef EC:1/1 62.5kHz HNS Head/FL:s FOV:22x22/2 3,0thk/1.5sp 36/03:52 612x512/1.50 NEX TRF PIR W = 7067 L = 4193 EFFR:0.43
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Landmarking:



Best Use:	Brain
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HNS Neck

Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Face, 1, 2, and 2 Chest elements (30cm FoV)

	Example Image 
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Landmarking:

Best Use:	Smallest FoV Cervical spine, Carotids, Soft Tissue neck.
------------------	--

HNS NeckChest

Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Face, 1, 2, and 5 Chest elements (35cm FoV)

	Example Image  PRFSE-XL/90 TR:3000 TE:108.4/8f EC:1 /1 /1 31.2kHz HNS NeckChest FOV:33x33cm 5.0mm/5.0mm 12/02/18 320x224/4.00 NEX FCI/NP/ED/TRF/SPF
--	--

Landmarking:



Best Use:	Large FoV Cervical spine, Carotids, Soft Tissue neck.
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HNS NVFull

Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Posterior Head, Face, 1, 2, and all 5 Chest elements (48cm FoV)

	Example Image 
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Landmarking:

Best Use:	Largest FoV Carotids, Cervical spine and Soft Tissue neck.
------------------	--

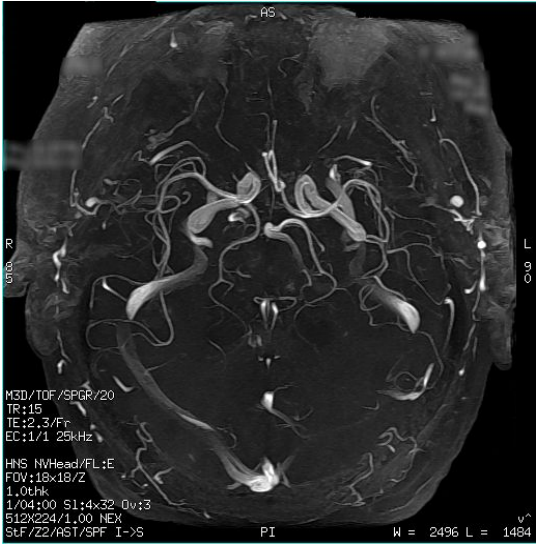
HNS NVHead

Components:	Posterior Head, Face and Chest (with or without TL)
Element Groups:	Posterior Head, Face and 1 (30cm FoV)



WW

Example Image



Landmarking:

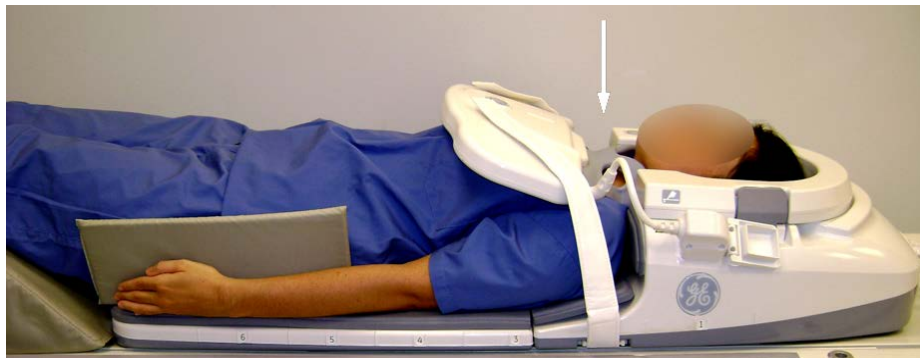


Best Use:	Brain
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HNS BrachPlex

Components:	Posterior head, Horseshoe, Chest and TL
Element Groups:	1, 2, 3 and all 5 Chest elements (40cm FoV)

	Example Image 
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Landmarking:

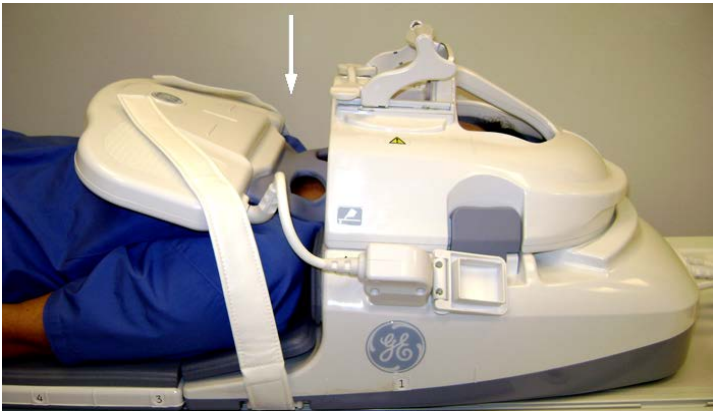
Best Use:	Brachial Plexus, Cervical spine.
------------------	----------------------------------

HNS BrachPlx

Components:	Posterior Head, Face, Chest, and TL
Element Groups:	1, 2, 3 and all 5 Chest elements (40cm FoV)

	Example Image 
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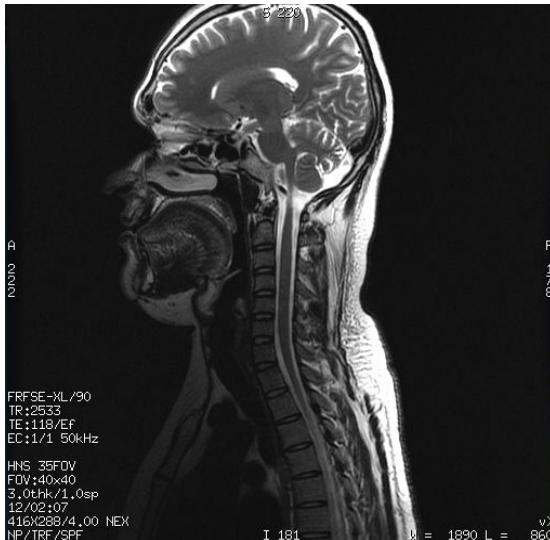
Landmarking:



Best Use:	Brachial Plexus, Cervical Spine.
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HNS 35FOV

Components:	Posterior Head, Face and TL
Element Groups:	Posterior Head, Face, 1, 2, and 3 (35-40cm FoV)

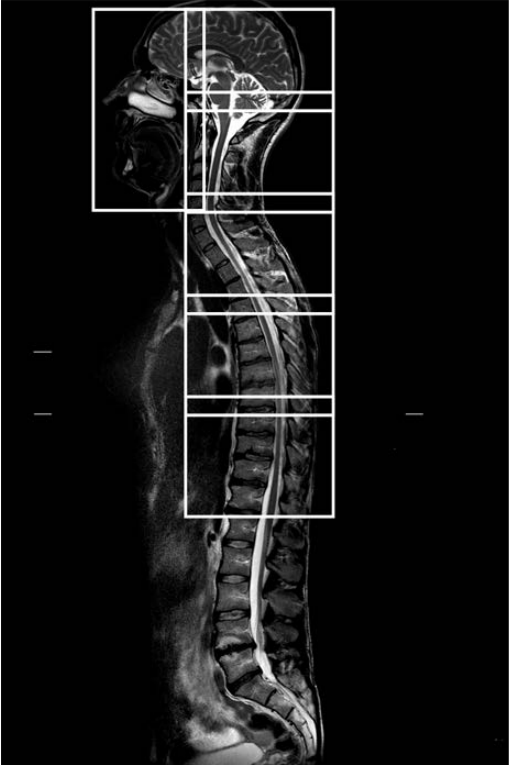
 WW: 175	Example Image  FRFSE -XL/90 TR:2533 TE:118/EF EC:1/1 50kHz HNS 35FOV FOV:40x40 3.0tHx/1.0sp 12/02:07 416X288/4.00 NEX NP/TRF/SPF I 181 X = 1890 L = 860
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Landmarking:

Best Use:	Large FoV Brain, Cervical Spine.
------------------	----------------------------------

HNS 48FOV

Components:	Posterior Head, Face and TL
Element Groups:	Posterior Head, Face, 1, 2, 3, and 4 (40-48cm FoV)

	Example Image  SR A 2 7 4 P 2 0 6 V-A IL FSE-XL90 TR:600 TE:13.1/Ef EC:1 /1 31.2kHz HNS 48FOV FOV:48x48/W 5.0thk/3.0sp 11/02/56 512x288/3.00 NEX TRF
--	---

Landmarking:



Best Use:	Largest FoV Brain, Cervical spine, Thoracic spine.
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Using the Equivalent Spine Configurations

The previous Section, “Using Configurations,” only shows the scan image, selected element groups and landmark position for the 12 spine configurations that are available when using Posterior Head, Horseshoe and TL spine. For convenience, each spine configuration has 3 equivalent configurations with other coil components connected. The following table shows the equivalent spine configurations. These equivalent configurations use the same element groups and have the same coverage. Refer to the previous section for the landmark information for these equivalent spine configurations.

Table 3-3: Equivalent Spine Configurations

Spine Configurations With Posterior Head and Horseshoe	Equivalent Spine Configurations		
	Posterior Head and Face	Posterior Head, Face and Chest	Posterior Head, Horseshoe and Chest
HNS CTL12	HNSp CTL12	HNS-CTL12	HNS_CTL12
Spine Configurations With Posterior Head, Horseshoe and TL Spine	Equivalent Spine Configurations		
	Posterior Head, Face and TL Spine	Posterior Head, Face, Chest and TL Spine	Posterior Head, Horseshoe, Chest and TL Spine
HNS CTL23	HNSp CTL23	HNS-CTL23	HNS_CTL23
HNS CTL123	HNSp CTL123	HNS-CTL123	HNS_CTL123
HNS CTL234	HNSp CTL234	HNS-CTL234	HNS_CTL234
HNS CTLTOP	HNSp CTLTOP	HNS-CTLTOP	HNS_CTLTOP
HNS CTLMID	HNSp CTLMID	HNS-CTLMID	HNS_CTLMID

For Users of the MR750w and Newer Systems

The following section is the Coil Mode Configuration and a matrix that explains the modes for using the HNS coil with the MR750w and newer systems. The matrix also contains the HNS configuration differences between the MR750w and the older systems such as the MR750, MR450, MR450w. Please refer back to Figure 3-15 for the Element Group name and number locations, if needed.

Table 3-4: Coil Configuration Matrix

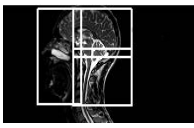
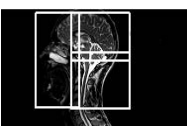
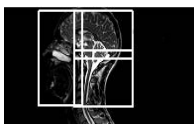
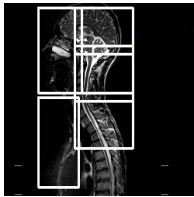
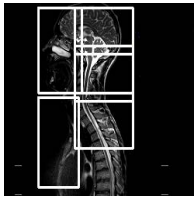
	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS Head/Neck/ Chest	No Coil Mode	No Coil Mode	HNS by GE (Posterior Head, Face)	HNS by GE (Posterior Head, Face)
Description	HNS Head			HNS Head	HNS NVHead
Coil Setting and Landmarking	Please use Coil Configuration 3 or Coil Configuration 4				
Scan Area (Face, Posterior Head, Group 1)					
Coils	HNS Head/Neck/ Chest	No Coil Mode	No Coil Mode	No Coil Mode	HNS by GE (Posterior Head, Face, Chest)
Description	HNS NVfull				HNS NVfull
Coil Setting and Landmarking	Please use Coil Configuration 4				
Scan Area (Face, Chest, Posterior Head, Groups 1, 2)					

Table 3-4: Coil Configuration Matrix (Continued)

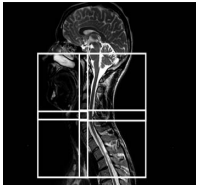
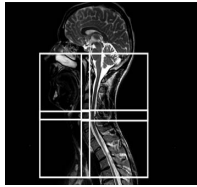
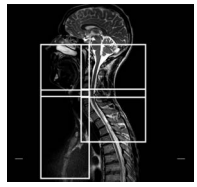
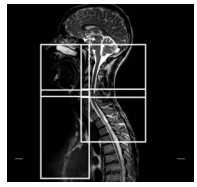
	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS Head/Neck/ Chest	No Coil Mode	No Coil Mode	No Coil Mode	HNS by GE (Posterior Head, Face, Chest)
Description	HNS Neck				HNS Neck
Coil Setting and Landmarking	Please use Coil Configuration 4				
Scan Area (Face, Chest, Groups 1, 2)					
Coils	HNS Head/Neck/ Chest	No Coil Mode	No Coil Mode	No Coil Mode	HNS by GE (Posterior Head, Face, Chest)
Description	HNS NeckChest				HNS NeckChest
Coil Setting and Landmarking	Please use Coil Configuration 4				
Scan Area (Face, Chest, Groups 1, 2)					

Table 3-4: Coil Configuration Matrix (Continued)

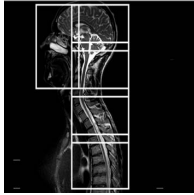
	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS Head/Neck/ Chest or HNS CTL Spine	No Coil Mode	No Coil Mode	HNS by GE (Posterior Head, Face, TL)	No Coil Mode
Description	HNS 35FOV			HNS 35FOV	
Coil Setting and Landmarking	Please use Coil Configuration 3				
Scan Area (Face, Posterior Head, Groups 1, 2, 3)					
Coils	HNS Head/Neck/ Chest or HNS CTL Spine	No Coil Mode	No Coil Mode	HNS by GE (Posterior Head, Face, TL)	No Coil Mode
Description	HNS 48FOV			HNS 48FOV	
Coil Setting and Landmarking	Please use Coil Configuration 3				
Scan Area (Face, Posterior Head, Groups 1, 2, 3, 4)					

Table 3-4: Coil Configuration Matrix (Continued)

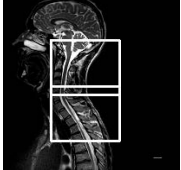
	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS Head/Neck/ Chest or HNS CTL Spine	No Coil Mode	HNS by GE (Posterior Head, Horseshoe, Chest, TL)	No Coil Mode	HNS by GE (Posterior Head, Face, Chest, TL)
Description	HNS BrachPlex		HNS BrachPlex		HNS BrachPlx
Coil Setting and Landmarking	Please use Coil Configuration 2 or Coil Configuration 4				
Scan Area (Chest, Groups 1, 2, 3)					
Coils	HNS CTL Spine	HNS by GE (Posterior Head, Horseshoe)	HNS by GE (Posterior Head, Horseshoe, Chest)	HNS by GE (Posterior Head, Face)	HNS by GE (Posterior Head, Face, Chest)
Description	HNS CTL12	HNS CTL12	HNS_CTL12	HNSp CTL12	HNS-CTL12
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 1, 2)					

Table 3-4: Coil Configuration Matrix (Continued)

	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS CTL Spine	HNS by GE (Posterior Head, Horseshoe, TL)	HNS by GE (Posterior Head, Horseshoe, Chest, TL)	HNS by GE (Posterior Head, Face, TL)	HNS by GE (Posterior Head, Face, Chest, TL)
Description	HNS CTL123	HNS CTL123	HNS_CTL123	HNSp CTL123	HNS-CTL123
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 1, 2, 3)					
Coils	HNS CTL Spine	HNS by GE (Posterior Head, Horseshoe, TL)	HNS by GE (Posterior Head, Horseshoe, Chest, TL)	HNS by GE (Posterior Head, Face, TL)	HNS by GE (Posterior Head, Face, Chest, TL)
Description	HNS CTL234	HNS CTL234	HNS_CTL234	HNSp CTL234	HNS-CTL234
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 2, 3, 4)					

Table 3-4: Coil Configuration Matrix (Continued)

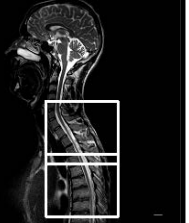
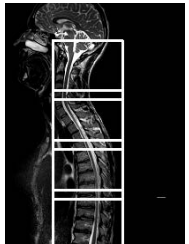
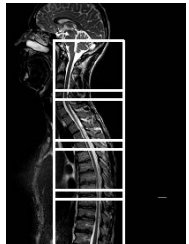
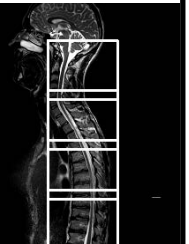
	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS CTL Spine	HNS by GE (Posterior Head, Horseshoe, TL)	HNS by GE (Posterior Head, Horseshoe, Chest, TL)	HNS by GE (Posterior Head, Face, TL)	HNS by GE (Posterior Head, Face, Chest, TL)
Description	HNS CTL23	HNS CTL23	HNS_CTL23	HNSp CTL23	HNS-CTL23
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 2, 3)					
Coils	HNS CTL Spine	HNS by GE (Posterior Head, Horseshoe, TL)	HNS by GE (Posterior Head, Horseshoe, Chest, TL)	HNS by GE (Posterior Head, Face, TL)	HNS by GE (Posterior Head, Face, Chest, TL)
Description	HNS CTLTOP	HNS CTLTOP	HNS_CTLTOP	HNSp CTLTOP	HNS-CTLTOP
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 1, 2, 3, 4)					

Table 3-4: Coil Configuration Matrix (Continued)

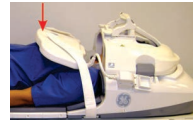
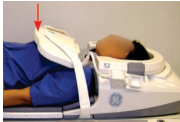
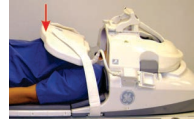
	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS CTL Spine	HNS by GE (Posterior Head, Horseshoe, TL)	HNS by GE (Posterior Head, Horseshoe, Chest, TL)	HNS by GE (Posterior Head, Face, TL)	HNS by GE (Posterior Head, Face, Chest, TL)
Description	HNS CTLMID	HNS CTLMID	HNS_CTLMID	HNSp CTLMID	HNS-CTLMID
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 2, 3, 4, 5)					
Coils	HNS CTL Spine	HNS by GE (TL (with or without Posterior Head and Horseshoe))	HNS by GE (TL (with or without Posterior Head, Horseshoe and Chest))	HNS by GE (TL (with or without Posterior Head, and Face))	HNS by GE (TL (with or without Posterior Head, Face, and Chest))
Description	HNS CTL34	HNS CTL34			
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 3, 4)					

Table 3-4: Coil Configuration Matrix (Continued)

	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS CTL Spine	HNS by GE (TL (with or without Posterior Head and Horseshoe))	HNS by GE (TL (with or without Posterior Head, Horseshoe and Chest))	HNS by GE (TL (with or without Posterior Head, and Face))	HNS by GE (TL (with or without Posterior Head, Face, and Chest))
Description	HNS CTL45	HNS CTL45			
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 4, 5)					
Coils	HNS CTL Spine	HNS by GE (TL (with or without Posterior Head and Horseshoe))	HNS by GE (TL (with or without Posterior Head, Horseshoe and Chest))	HNS by GE (TL (with or without Posterior Head, and Face))	HNS by GE (TL (with or without Posterior Head, Face, and Chest))
Description	HNS CTL56	HNS CTL56			
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 5, 6)					

Table 3-4: Coil Configuration Matrix (Continued)

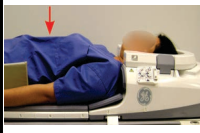
	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS CTL Spine	HNS by GE (TL (with or without Posterior Head and Horseshoe))	HNS by GE (TL (with or without Posterior Head, Horseshoe and Chest))	HNS by GE (TL (with or without Posterior Head, and Face))	HNS by GE (TL (with or without Posterior Head, Face, and Chest))
Description	HNS CTL345	HNS CTL345			
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 3, 4, 5)					
Coils	HNS CTL Spine	HNS by GE (TL (with or without Posterior Head and Horseshoe))	HNS by GE (TL (with or without Posterior Head, Horseshoe and Chest))	HNS by GE (TL (with or without Posterior Head, and Face))	HNS by GE (TL (with or without Posterior Head, Face, and Chest))
Description	HNS CTL456	HNS CTL456			
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 4, 5, 6)					

Table 3-4: Coil Configuration Matrix (Continued)

	MR750w	Coil Configuration			
		Configuration 1 - Posterior Head, Horseshoe	Configuration 2 - Posterior Head, Horseshoe, Chest	Configuration 3 - Posterior Head, Face	Configuration 4 - Posterior Head, Face, Chest
Coils	HNS CTL Spine	HNS by GE (TL (with or without Posterior Head and Horseshoe))	HNS by GE (TL (with or without Posterior Head, Horseshoe and Chest))	HNS by GE (TL (with or without Posterior Head, and Face))	HNS by GE (TL (with or without Posterior Head, Face, and Chest))
Description	HNS CTLBOT	HNS CTLBOT			
Coil Setting and Landmarking	Please use Coil Configuration 1, 2, 3 or 4				
Scan Area (Groups 3, 4, 5, 6)					

Coil and Patient Positioning



WARNING:

RF coupling danger. Patient burns may result from RF coupling of this device with other devices remaining in the magnet. Remove any other coil or unused accessory device from the magnet before using this coil.



WARNING:

Patient burn risk. Do not cross or loop coil cables. Do not allow any cable to touch the patient. Use a thermal-resistant material or pad to prevent cables from touching the patient.



CAUTION:

Be sure the coil is unplugged before lowering the table. Failure to do so may damage the coil, cable, or connector. The coil may not work if damaged.

NOTE: For ease of use, position the patient restraint straps in the desired location on the table before placing the TL Component on the table.

Landmarks are indicated for each configuration. Use these as a guideline when positioning your patients.

The raised laser mark on the Anterior Face component is used for landmarking. The TL Spine component has raised laser marks positioned between the numbered groups identifying the beginning and end of each element group.

**CAUTION:**

Ensure that no hair or fabric is caught between the components. If the components are not properly connected, oscillation artifacts will appear on the image. If artifacts occur, reseal the components and repeat the scan.

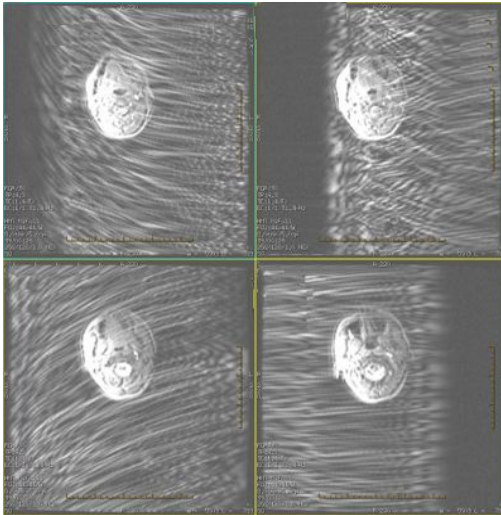


Figure 3-16: Oscillation artifact

**CAUTION:**

This label, affixed to the Anterior Face, Horseshoe and Chest connector informs the user to firmly press down on all coil components to ensure that they are fully engaged and connected. Failure to do so may cause artifacts and decrease image quality. Refer to Figure 3-16.

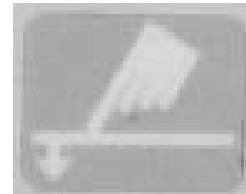


Table 3-5: Coil and Patient Positioning

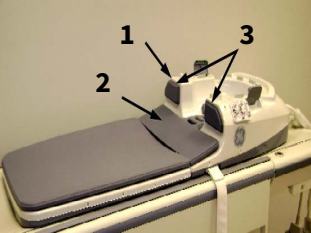
Visual	Description
	1. Place the TL Spine component (arrow) on the table, then place the pad on it. Patient will enter the scanner head first, so be sure cables extend toward the docking end of the table.
	2. Place the Posterior Head component (arrow 1) on the table as shown, making sure it clears the cradle docking mechanism. Interlock the pegs on the bottom with the holes in the TL Spine Component. Place pad on the Posterior Head component as shown (arrow 2). Place the patient pads on the posterior Head component as shown (arrow 3). NOTE: Ensure that the TL cable lies within the cable channel beneath the Posterior Head component. When placed properly, the Posterior Head component will sit evenly on top of the TL cable.
	3. Position the patient in the supine position on the table and lower the head into the Posterior Head component. Make sure that shoulders touch the shoulder rest area (arrow).
	4. Place the Face component (arrow 1) over the Posterior Head component. Secure Face component with the blue latches on both sides (arrow 2).  CAUTION: Ensure that no hair or fabric is caught between the components. Failure to comply may cause artifacts and decreased image quality.

Table 3-5: Coil and Patient Positioning (Continued)

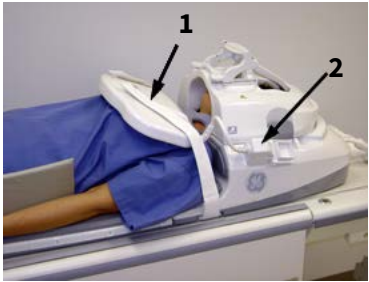
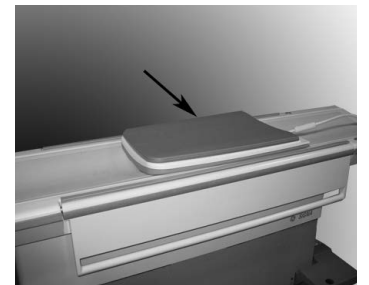
Visual	Description
	5. Position the Chest component (arrow 1) so that the neck portion fits snugly against the patient's chin. Plug the chest piece connector into the Posterior Head component (arrow 2).  CAUTION: Ensure that the Chest component is plugged securely into the connector on the Posterior Head component. Failure to comply may cause artifacts and decreased image quality.  CAUTION: The Chest component may not fit in the bore on patients with large chests. To avoid injuring the patient, watch carefully as the table moves into the bore, and stop advancing the table if the Chest component comes into contact with the top of the bore.
	6. When scanning only C, T or L spine, it is not necessary to attach the Face component. Instead, the Horseshoe (arrow) may be used. It is positioned much like the Face component and is secured with the blue latches on either side of the Posterior Head component. NOTE: The Face component or the Horseshoe MUST be in place for the coil to function properly when the Posterior Head component is used.  CAUTION: Ensure that the Horseshoe component is completely seated on the Posterior Head component. Failure to comply may cause artifacts and decreased image quality.
	7. If using only the TL Spine component of the array, place TL component on the table, making sure cable extends to docking end of table as shown.

Table 3-5: Coil and Patient Positioning (Continued)

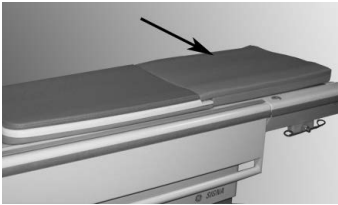
Visual	Description
	8. Place patient comfort pad over cable as shown. Make sure that cable follows the groove in the underside of the pad. 9. Position patient and then plug the TL connector into port A. If you are using only the TL Spine component, you may plug the TL connector into port B. In general, always use port A. NOTE: The patient may be positioned Head first or Feet first when using the TL Spine component independently.

Table 3-5: Coil and Patient Positioning (Continued)

Visual	Description
 HD Connector	10. There are multiple system coil plug-in configurations. - For a three connector HD coil - Plug the A connector into the A port, the B connector into the B port and the Legacy Bendix into the Bendix port for a 1.5T System or the 3.0T Legacy Connector into the Legacy port for a 3.0T System. - For a two connector HD Coil - Plug the A connector into the A port and the B connector into the B port. - For a P connector coil - Plug the A connector into the A port and the P connector into either Port P1 or P2. Lock the P connector plug by rotating the spindle handle until the Lock symbol is visible. NOTE: Make sure all cables are plugged in and that they do not cross or loop. When cables labeled A or B are plugged in, the green light illuminates.
 P connector	
 Legacy Bendix connector	 CAUTION: All coil components must be plugged into the system when they are in the scanner. Leaving components unplugged can damage the coil, or cause harm to the patient.
 3.0T Legacy Connector	11. Disconnect coil cables before lowering the patient table.  CAUTION: Unplug the coil before lowering the table. Failure to do so may cause damage to the coil and connector and may render the coil inoperable.

Localizing anatomy

Position the area of interest to the center of the coil elements (based on selected configuration). Please refer to Table 3-1 or 3-4 for detailed coil configurations. Landmark to the center of the coil elements at the region of interest.

Recommended scan protocols

Use the system recommended protocols for scanning with this coil for optimum results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

Workflow

To best utilize the HNS Coil, perform the following steps:

1. Set up the coil components necessary to scan the region of interest.

NOTE: Ensure that all the components on the table are plugged in before scanning.

2. Register the patient then select Coil Type “Currently Connected.”
3. Select the best configuration for the region of interest (keep in mind the maximum FoVs for the selected configuration).

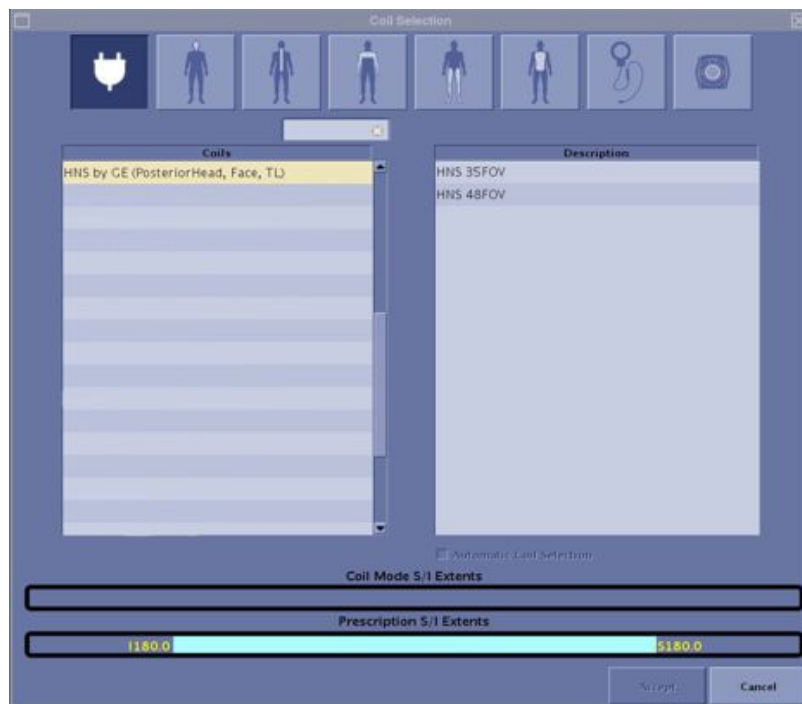


Figure 3-17: Coil Selection window

NOTE: Some configurations are only compatible when certain coil components are plugged into the system. If the configuration you want is not listed, it may be necessary to add or remove coil components before your configuration is selectable. (Refer to Table 3-1 or 3-4 for a list of configurations and compatible components.)

4. Position the area of interest over the elements in the selected configuration.
5. Landmark to the middle of the element groups used in the selected configuration, then Advance to Scan.

NOTE: Precise patient positioning and proper FoV selection will yield optimum results.
Improper use of the coil is the major cause of image artifacts. To diminish artifacts, it is important to place the body part to be scanned over the elements in the selected configuration then landmark to those elements. Refer to the section coil mode configuration sections starting on page 3-13 and page 3-44 for guidance.

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Chapter 4

Coil Maintenance

Introduction - Coil Maintenance

To maintain the highest possible image quality, and to protect your patients and yourself from potential injury or illness, it is important to keep the coil clean and well maintained. In this chapter, we discuss basic methods for keeping the coil safe and sanitary. We also provide you with a list of accessories that will enhance the utility of your coil, and you will find contact information in this section should you have questions or comments you would like to pass along.

Quality Assurance

Coil Quality Assurance tests must be run by a GE Service Representative or a third party service provider. To have a Quality Assurance test run on a coil, please call your GE Service Representative or your third party service provider.

If you do not have a GE Service Representative or third party service provider, use the Class A Service CD that was shipped with your MR System. This CD contains information on phantom setup and the usage of the MCQA tool. Please contact GE with any questions or concerns.

Coil Maintenance

The coil must be inspected before each use. Visually check each coil component, cable and accessory to ensure that there is no external damage. If any coil component, cable or accessory is suspected of not being in good condition, discontinue use and contact a GE Service Engineer. Please contact a service engineer if:

- The coil develops any tears or is damaged in a way that exposes internal padding or conductors.
- The coil becomes severely soiled beyond what can be cleaned according to the instructions found later in this chapter.
- There are image artifacts seen during scanning.

All FRU's (Field Replaceable Units) must be replaced by a GE Service Engineer or a third party service provider. Component part lists with descriptions, instructions for the safe replacement of components, circuit diagrams, and calibration and preventive maintenance instructions are available to service personnel upon request.

General Cleaning and Disinfection



CAUTION:

To avoid possible damage to equipment, do not use solutions containing amines, strong alkalis, quaternary ammonium chloride compounds, esters, iodine, aromatic or chlorinated hydrocarbons, or ketones. Do not use autoclaves or the industrial washers and dryers found in most hospitals or professional laundry services.



WARNING:

Electric shock may occur if the coil is attached to the system during cleaning or when it is still wet. Detach coil connector from the scanner before attempting to clean the coil. Do not touch connectors with bare fingers. Never press sharp objects against connector surface. Do not reattach connector after cleaning until the coil has dried completely.



CAUTION:

The coil contains sensitive electronic components that may be damaged by liquids. Do not spray or pour cleaning solution directly onto the coil. Do not submerge the coil in any solution. Under no circumstances should the coil be placed into any type of sterilizer.

Clean and disinfect the device after removing from packaging prior to first use. Then, cleaning and disinfection of patient contact surfaces is recommended following each use.

Non-patient contact surface cleaning and disinfection should be conducted following internal housekeeping procedures unless otherwise indicated in the following instructions.



CAUTION:



Inspect pads for peeling or cracking. To prevent a biohazard, replace cracked or peeling pads before using.

Cleaning and Disinfection Recommendations for the MR System, Coils and Most Accessories

- Clean with commercially available wipes that contain 0.525% minimum sodium hypochlorite as the only active ingredient. If commercially available wipes are not available, then follow one of these instructions.
 - Clean with a lint free cloth saturated with a 1:10 dilution of commonly available bleach containing a recommended minimum sodium hypochlorite of 5.25%. Dilute the bleach with tap water.
 - Use a lint free cloth that has been saturated with a 70% isopropyl alcohol solution made from 100% isopropyl alcohol and 30% tap water.
 - Use a lint free cloth with a commercially prepared solution of 70% concentration isopropyl alcohol.
 - Carefully clean all surfaces including cracks and crevices. Consider placing a wipe around a rigid implement (GE Part Number 5878912 or equivalent) and use this to get into cracks. Use swabs covered with wipes or diluted cleaning agents to access blind holes and recessed areas.
-

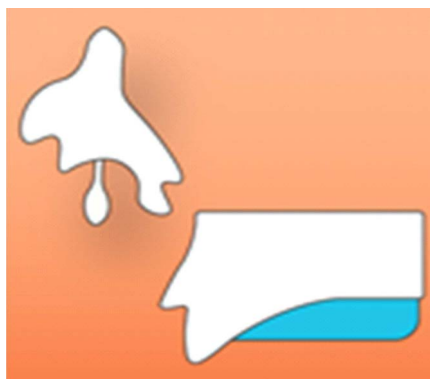


Figure 4-1: Suggestions for rigid cleaning implements. The upper left image is a swab covered with a wipe. The lower right image is of a card covered with a wipe.



Figure 4-2: Example of a cleaning tool

- Regardless of the method used, inspect to ensure visual cleanliness prior to disinfection. Repeat the cleaning process until all visible soil has been removed.

- Disinfect with commercially available wipes that contain 0.525% minimum sodium hypochlorite as the only active ingredient following the manufacturer's instructions. If commercially available wipes are not available, then follow these instructions.
 - Disinfect with a lint free cloth and a 1:10 dilution of commonly available bleach containing a recommended minimum sodium hypochlorite of 5.25%. Dilute the bleach with tap water.
 - For general disinfection or disinfection following cleaning of blood and/or body fluids, 5 minutes contact time is recommended.
 - Use the implements described above for disinfection, if necessary.
 - Refer to internal procedures or refer to publications such as "CDC Guideline for Disinfection and Sterilization in Healthcare Facilities," 2008 or latest revision, for guidance. Disinfectant may need to be reapplied to ensure surfaces remain wet for the duration of the selected contact time.
 - After you have cleaned and disinfected with bleach (sodium hypochlorite), wipe surfaces with a disposable lint free wipe that has been dampened with purified water to remove any remaining bleach residue and allow device to air dry indoors before use.
- Should your coil need to be returned to GE for service, please clean and disinfect the component being returned as instructed above.
- Dispose of any materials used to clean the coil and the pads according to all federal, state, and local regulations.

Accessories and Replaceable Components

Coil Replacement parts (FRU's - Field Replaceable Units) should be ordered by a GE Service Representative or a third party service provider.

If you do not have a GE Service Representative or third party service provider, please contact GE.

Table 4-1: 1.5T and 3.0T HNS Coil Accessories

Part Name	Part Number
1.5T and 3T MR Head/Neck/Spine (HNS) Coil Pad Set	E8801WG
Phantom Positioner	5344844
FRU, 1.5T HNS, 3T HNS, and 3T NV Hardware Kit	5365916

Glossary

C	Celsius
CD	Compact Disc
CDC	Centers for Disease Control and Prevention - the national health agency of the United States
cm	Centimeter
DVD	Digital Video Disc
ECG	Electrocardiogram
eIFU	Electronic Instructions for Use
EMC	Electromagnetic Compatibility
F	Fahrenheit
FOV or FoV	Field of View
FRU	Field Replaceable Unit
HNS	Head, Neck, and Spine
HNU	Head Neck Unite
hPa	A unit of pressure; Hecto Pascal
in	Inch
IP Rating	Ingress Protection rating. The effectiveness of the seal of the coil against intrusion from foreign bodies and moisture
kg	Kilogram
lbs	Pounds
LPCA	Low Profile Carriage Assembly
MCQA	Multicoil Quality Assurance

MHz	Megahertz
mm	Millimeters
mm Hg	A unit of pressure; millimeters of mercury
MRI	Magnetic Resonance Imaging - The electromagnetic signal (in the radio frequency range) produced by the precession of the transverse magnetization of the spins. The rotation of the transverse magnetization induces a voltage in the coil. This voltage is amplified by the receiver.
NCU	Neck Chest Unit
QA	Quality Assurance
RF	Radio Frequency - The frequency (intermediate between audio and infrared frequencies) used in magnetic resonance systems to excite nuclei to resonance.
TLU	Thoracic Lumbar Unit. This is also referred to as the TL Spine or simply TL in this manual
Type BF	The rating of protection against electrical shock for the coil